

CS304 Quiz #1

CS304 Quiz #1 by

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1. Suppose derived class is inherited from base class. What happens when a derived class object is created?

The constructor of base class is executed before the constructor of derived class.

2. Suppose str1, str2 and str3 are objects of class String. Choose appropriate declaration of overloaded assignment operator for the following statement to work correctly.
str1 = str2 = str3;

String& operator =(const String &);

3. In C++, which of the following is defined as stream insertion operator?

<<

4. Choose correct declaration of overloaded inequality (!=) operator for class String as non-member friend function.

None of the given options

5. Relationship in which child object gets destroyed if parent object is destroyed.

Composition

6. How the information hidden within an object can be accessed?

Through its interface

7. Which of the following defines default constructor for class "Student"?

Student() {//...

}

8. The sub-object's life is not dependent on the life of master class in _____.

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Aggregation

9. Hiding the implementation details makes program:

Easy to understand

10. The functions defined inside the class are by default _____.

Inline

11. Which of the following statement is TRUE about static function of a class?

It is used to access static data members.

12. Deconstruct is a function which has the same name as that of class but starts with a—
sign

~

13. 'this' pointer cannot be passed implicitly to _____ functions.

Static Member

14. Which of the following can be the behavior of an object "Usman"?

Eat

15. In Object Oriented programming, objects communicate with each other through

Messages

16. Constructor is used to _____ the objects of a class.

Initialize

17. Which of the following is NOT an access specifier in C++?

Hidden

18. The other name of subtyping is _____.

Extension

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19. Suppose a class does not have any constructor. What will happen when an object of this class is created?

Compiler will call implicit default constructor.

20. In case when we define the function outside the class then we must use the keyword _____ to make the function inline.

Inline

21. In constant member function the type of this pointer is:

Constant pointer to constant data

22. In _____, base class can be replaced by the derived class.

Extension

23. We can get the address of a variable stored in a pointer using _____:

& symbol

24. A child inherits characteristics from its _____.

Parent

25. Which of the following function declaration is correct to overload the + operator as member function in Complex class?

Complex operator +(const Complex & rhs);

26. Which of the following is not true about constant member function?

All of the given options

27. What is the general syntax of overloading Unary Operator as member function of the class?

TYPE & operator OP ();

28. _____ is creating objects of one class inside another class.

Composition

29. Which of the following is true about constructor?

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All of the given option

30. Memory is allocated to non static members only, when

Object is created

31. A child inherit characteristics from its ?

parents

32. A post-fix unary operator is implemented in C++ member function with

1 dummy int arguments

33. Which of the following is the example of objects in school?

Books , pages

34. Suppose student is a class, which of the following constructor with one parameter for class student?

Student()

35. If you have three classes in a C++ program A, B, and C where class A inherits from class B, then class _____ contains all the characteristics of class _____.

A, C

36. In ____ base class can be replaced by the derived class.

Extension

37. The concept of derived class is involved in _____

inheritance

38. Suppose person is a class which of the following statement defines an object of class person?

Create person object

39. Which of the following is true about Accessor Functions?

a) They are also used to access private data of the object.

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- b) It can be used to set private data member of the class
 - c) It can be used to get private data member of the class
 - d) All of the given options.
40. The other name of subtyping is
Extension
41. Which of the following statement best describes the constructor
constructor is used to initialize the data member of a class.
42. Which of the following is true about inline function?
It is used by compilers to improve efficiency of the program
43. Which of the following is defined as stream extraction operator in c++?
>>
44. An object has ____ interface(s)
One or more than one
45. Which of the following is an advantage of encapsulation?
Better understanding
46. Which of the following represents the two-way association?
Employee works for the company
47. ____ is the weakest link between object.
Simple association
48. Which of the following statement is NOT true about static variable of a class?
Static variable belongs to a particular instance of a class
49. In composition _____ are called from composing objects to compound objects.
Constructor
50. In OOP we can achieve reusability though _____
Inheritance
51. Generalization is implemented through
Inheritance

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52. Suppose str1, str2, and str3 are objects of class string. Choose appropriate declaration of overloaded assignment operator for the following statement to work correctly.

`String& operator=(const String &)`

53. In C++ which of the following is defined as stream insertion operator.

`<<`

54. Abstraction hides the ____ details.

Relevant

55. Minimum classes required in a program for implementing multiple inheritance.

3

56. Composition is _____ and aggregation is _____ relationship.

Strong, weak

57. Minimum classes required for implementing multiple inheritance.

At least 3

58. We can access a private static variable through.

Static data member

59. The process of extracting common characteristics from two or more classes and combining them into a generalized superclass is called ____

Generalization

60. Which of the following is true about polymorphism?

All of the given

61. Which of the following can be the behavior of an object "Usman"?

Eat

62. Composition is ____ relationship

Part whole

63. Which of the following is a strong relationship?

Composition

64. Which of the following defines default constructor for class student?

```
Student(){...  
}
```

65. This car is composed of wheels. What is the relationship between car and wheels?

Composition

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66. In ____ base class can be replaced by the derived class.

Extension

67. ____ - ____ defines the order of evaluation of an operator in an expression.

Operator precedence

68. In composition ____ are called from composing objects to composed objects.

Destructor

69. How many objects of a given class may be constructed in an application?

As many as the application asks for.

70. Relationship in which child object gets destroyed, If parent object is destroyed.

Composition

71. Which of the following is not true about constructor?

Constructor is a special function which can have different name than class name.

72. Suppose test is a class, void show() is its public member function. which of the following is correct call of show() function using Test pointer ptr?

ptr->show()

73. Which of the following features of OOP is used to deal with only relevant details?

Abstraction

74. In C++ the subscript operator must be overloaded as a ____ of the class, with one parameter of ____ type.

Member function, int

75. Identify the abstract class from the given statement

“vehicle class is base class. Bus, Car and truck are derived classes”

Vehicle

76. How can we differentiate between constructor and destructors?

Destructor are preceded by a tilde(~) symbol and constructor are not preceded with any symbol.

77. If you do not initialize static variable of int type then it is automatically initialized with

0

78. Choose the correct statement

All of the given options

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79. ____ associates the objects of the exact two classes.

Binary association

80. Which of the following is the advantage of inheritance?

Reusability

81. Member function defined inside a class declaration are ____ by default

Public

82. In Object Oriented programming objects communicate with each other through ____

Messages

83. ____ associates the objects of the exact three classes.

Ternary Association

84. An objects has attributes, operations and ____

Unique identity

85. In C++ which of the following keywords works only with constructors?

Explicit

86. Which of the following allows to reuse characteristics of more than one parent class?

Multiple Inheritance

87. How many objects are involved in the Binary association?

2

88. Which of the following features of OOP is used to deal with only relevant details?

Abstraction

89. The overloaded '-' operator for complex class will be called with reference to ____ in the following statement

Complex C3 = C1 - C2

Complex

90. Static data member is declared ____

Inside the class

91. Class is blue print of ____

Object

92. Which of the following is true about inline functions?

All of the given options

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93. How many objects can be created once a class is declared?

As many as needed

94. Suppose you have following C++ statements

```
int oldValue=10
```

```
intviewValue * ++ old/Value
```

What will be the value of old/Value and new/Value after executing above statement?

old/Value=11, new/Value=11

95. In ____ base class can't always be replaced by the derived class.

Specialization

96. Which of the following will happen when a constant function tries to change the value of data members of the class?

Complex time error will occur

97. ____ is set of functions of an objects exposed to other objects.

Interface

98. How many objects are involved in the Ternary association?

3

99. Mermaid is an example of

Multiple inheritance

100. Which of the following is the correct syntax of declaring static variable 'count' of type int?

static int count

101. Which of the following operators operate on one operand?

Unary Operators

102. Object Data includes ____

Both static and attributes of an objects

103. In case when we define the function outside the class then we must use the keyword ____ to make function inline.

Inline

104. A function call is resolved at run time in

a) Non virtual members function

b) Virtual member function

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- c) Both none-virtual and virtual member function
- d) None of given

105. If the user does not specify the type of inheritance, then the default type of inheritance is

- a) Public inheritance
- b) Private inheritance
- c) Protected inheritance
- d) None of given

106. In case of private inheritance, private members of base class will be in derived class?

- a) Private
- b) Public
- c) Protected
- d) Hidden

107. Then int member i of base class is accessible in class,

- a) derived 1 only
- b) derived 2 only
- c) both derived 1 derived 2
- d) none of the given options

108. In case of public inheritance, protected members of base class will be in derived class?

- a) Private
- b) Public
- c) Protected
- d) Hidden

109. By default assignment operator (=) performs,

- a) Shallow copy
- b) Deep copy
- c) Both of these

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d) None of these

110. In protected inheritance the public members of base class become _ in derived class.

a) Public

b) Private

c) Protected

d) None of the given

111. Compiler can automatically generates the following constructor/s for a class,

a) Default constructor

b) Copy constructor

c) Both of these

d) None of these

112. Friend's functions of class are _ members of that class.

a) Public

b) Private

c) Protected

d) None of the given

113. Methodologies to the development of reusable software relate to .

a) Structure programming

b) Procedural programming

c) Generic programming

d) None of the given

114. Function overriding is done in context of,

a) Single class

b) Single derived class

c) Single base class

d) Derived and base classes

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115. Which of the following is not type of inheritance in c++?

- a. Public
- b. Protected
- c. Restricted
- d. Private

116. If there is a pointer ,p, to objects of a base class and it contains the address of an object of a derived class and both

classes contain a virtual member function, ding(), then the statement p-

>ding(); will cause the version of ding in the class to be executed.

- a) Base
- b) Derived
- c) Virtual
- d) Implemented

117. Target class of a function call is determined t run time.

- a. Instance
- b. Virtual
- c. Operator
- d. None of the given

118. To make a pure virtual, we need to give _after() of this function.

- a. =0;
- b. =1;
- c. Null;
- d. None of the given

119. If we have not given any constructor for the class, compiler

generates which of the following constructors?

- a. Explicit Default Constructor
- b. Implicit Default Constructor
- c. Explicit Parameterized Constructor
- d. Implicit Parameterized Constructor

120. In c++ by default access of classes is .

- a. Private

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- b. Protected
 - c. Public
 - d. None of the given
121. In public inheritance the public members of base class become in derived class
- a. Private
 - b. Protected
 - c. Public
 - d. None of the given
122. A class hierarchy
- a. Shows the same relationships as a organization chart
 - b. Describe “has a” relationship.
 - c. Describe “is a kind of” relationship.
 - d. Shows the same relationships as a family tree.
123. In private Inheritance the public members of the base class become in derived class
- a. Public
 - b. private
 - c. Protected
 - d. None of the given option
124. In the line No.2, which constructor of student class will be
- a. Default constructor of student class
 - b. Copy constructor of student class
 - c. Both Default and Copy constructor of student class
 - d. No constructor will be called
125. Which of the following is Not casting operator in c++ standard?
- a. Static-cast
 - b. Var-cast
 - c. Dynamic-cast
 - d. Reinterpret-cast

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126. Which of the following type of inheritance is used to model “Implemented in terms of” relationship?
- a. Public
 - b. Private
 - c. Protected
 - d. Confidential
127. When we want to implement one class in terms of another class then we use,
- a. Public inheritance
 - b. Protected inheritance
 - c. Private inheritance
 - d. None of these options
128. Public inheritance represents;
- a. “IS A “relationship
 - b. “Has A” relationship
 - c. “IS Special kind of” relationship
 - d. None of these option’
129. Suppose we have defined derived class copy constructor but have not defined base class copy constructor then compiler will,
- a. Use base class default constructor
 - b. Generate base class copy constructor itself
 - c. Use base class assignment operator
 - d. None of these
130. Child class can call constructor of its,
- a. Direct base class
 - b. Indirect base class
 - c. Both direct and indirect base classes
 - d. None of these
131. We can have type of member functions in a class.
- a. Public
 - b. Private
 - c. Protected
 - d. All of these options

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132. Compiler generated copy constructor performs
- a. Shallow copy
 - b. Deep copy
 - c. Both Shallow and Deep copy
 - d. None of these options
133. In c++ generic programming is done using
- a. Procedures
 - b. Packages
 - c. Templates
 - d. None of the given
134. Target class of a function call is determined at run time.
- a. Instance
 - b. Virtual
 - c. Operator
 - d. None of the given
135. Suppose person is a user defined class. In statement "Person*pPtr", static type of pPtr is ----.
- a. pPtr
 - b. pointer
 - c. person
 - d. None of the given
136. Which statement is true for concrete class?
- a. It cannot be inherit from an abstract class
 - b. It can be instantiated
 - c. It cannot be instantiated
 - d. None of given
137. In c++, compiler can generate which of the following operators 'code,
- a. ==
 - b. =
 - c. &
 - d. &&+
138. In specialization we can,
- a. Replace child class with its base class
 - b. Replace base class with its child class

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- c. Replace both child and base classes interchangeably
 - d. None of the given option
139. In private inheritance derived class pointer can be assigned to base class pointer in,
- a. Main function
 - b. In derived class member and friend class function
 - c. In base class member and friend functions
 - d. None of the given option
140. When we access private members of the class from outside the class using operator with its object pointer
- a. ->
 - b. .
 - c. &
 - d. None of above
141. Which of the following access specifier ensure that base class member is accessible in derived class and Not outside of this class
- a. Public
 - b. Private
 - c. Protected
 - d. All of above
142. Static casting is
- a. C++ way of calling base class function from derived class
 - b. C way of calling base class function from derived class
 - c. Both of these
 - d. None of these
143. Suppose both derive and base classes have compilers generated copy constructors then derived class copy construct is
- a. Compiler generated default constructor
 - b. Compiler generated copy constructor
 - c. Both of these
 - d. None of these
144. In case of protected inheritance, public members of the base class will be in derived class

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- a. Private
 - b. Public
 - c. Protected
 - d. Hidden
145. Methodologies to the development of reusable software related to
- a. Structured programming
 - b. Procedural programming
 - c. Generic programming
 - e. None of the given
146. A child class can call constructor of the parent call through
- a) Its constructor initialization list
 - b) Its constructor body
 - c) Both from its constructor initialization list or body
 - d) Cannot call the constructor of its parent class
147. In case of dynamic memory allocation in or class we should use
- a. Default assignment operator
 - b. User defined assignment operator
 - c. Default copy constructor
 - d. None of these
148. Consider the code below. Class class1{

Protected:

Int I;

};

Class class2: private calls 1{

};

- a) Public
- b) Protected
- c) Private

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d) None of the given options

149. Consider the following two lines of code written for class student

Student subj1;

Student subj2(subj1)

In line 2 which constructor of student class will be called?

- a) Default constructor of student class
- b) Copy constructor of student class**
- c) Both default and copy constructor of student class
- d) No constructor will be called

150. Which of the following is /are casting operator(s) inc++?

- a. Static_cast
- b. Const_cast
- c. Dynamic_cast
- d. All of the given options**

151. A template provides a convenient way to make a family of

- a. Variables and data members
- b. Functions and classes**
- c. Classes and exceptions
- d. Programs and algorithms

152. Consider the code below class class1{

Private:

Void func1();

};

Class class2 : public class1{

};

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Function func1 of class 1 is in class2.

- a) Public
- b) Protected
- c) Private
- d) None of the given options

153. We can call base class assignment operator in derived class user defined assignment operator

- a. Implicitly
- b. Explicitly
- c. Using both of these options
- d. Using none of these options

154. Function func1 of class1 is _ in class2.

- a. Public
- b. Protected
- c. Private
- d. None of the given

155. The default inheritance mode is,

- a. public inheritance
- b. Protected inheritance
- c. Private inheritance
- d. None of the given

156. Consider the code below class

```
class1{  
  
public:  
  
Void func1();  
  
};  
  
Class class2: protected class1{
```

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};

Function func1 of class 1 is in class2.

- a) Public
- b) Protected
- c) Private
- d) None of the given options

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- a) Variables and data members
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159. Then int member i of base class is accessible in class,

- a) derived 1 only
- b) derived 2 only
- c) both derived 1 derived 2
- d) none of the given options

160. In case of public inheritance, protected members of base class will be in derived class?

- a) Private
- b) Public
- c) Protected
- d) Hidden

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161. By default, assignment operator (=) performs,
- a) Shallow copy
 - b) Deep copy
 - c) Both of these
 - d) None of these
162. In protected inheritance the public members of base class become _ in derived class.
- a) Public
 - b) Private
 - c) Protected
 - d) None of the given
163. Compiler can automatically generate the following constructor/s for a class,
- a) Default constructor
 - b) Copy constructor
 - c) Both of these
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- a) Public
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- a) Structure programming
 - b) Procedural programming
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166. Function overriding is done in context of,
- a) Single class
 - b) Single derived class

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- c) Single base class
- d) Derived and base classes

167. Which of the following is not type of inheritance inc++?

- a) Public
- b) Protected
- c) Restricted
- d) Private

168. If there is a pointer, p, to objects of a base class and it contains the address of an object of a derived class and both classes contain a virtual member function, ding(), then the statement p->ding(); will cause the version of ding in the class to be executed.

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- a) Explicit Default Constructor
- b) Implicit Default Constructor

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- c) Explicit Parameterized Constructor
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172. In c++ by default access of classes is .

- a) Private
- b) Protected
- c) Public
- d) None of the given

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- c) Public
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- a) Shows the same relationships as a organization chart
- b) Describe “has a” relationship.
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- c) “IS Special kind of” relationship
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181. Suppose we have defined derived class copy constructor but have not defined base class copy constructor then compiler will,

- a) Use base class default constructor
- b) Generate base class copy constructor itself
- c) Use base class assignment operator
- d) None of these options is correct

182. Child class can call constructor of its

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- a) Direct base class
- b) Indirect base class
- c) Both direct and indirect base classes
- d) None of these

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- b) Private
- c) Protected
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186. Suppose person is a user defined class. In statement "Person*pPtr", static type of pPtr is ----.

- a) pPtr
- b) pointer
- c) person
- d) None of the given

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- a) It cannot be inherit from an abstract class
- b) It can be instantiated
- c) It cannot be instantiated
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188. In c++, compiler can generate which of the following operators 'code
- a) ==
 - b) =
 - c) &
189. In specialization we can,
- a) Replace child class with its base class
 - b) Replace base class with its child class
 - c) Replace both child and base classes interchangeably
 - d) None of the given option
190. In private inheritance derived class pointer can be assigned to base class pointer in,
- a) Main function
 - b) In derived class member and friend classfunction
 - c) In base class member and friend functions
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- a. ->
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- a. C++ way of calling base class function from derived class

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- b. C way of calling base class function from derived class
 - c. Both of these
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194. Suppose both derive and base classes have compilers generated copy constructors then derived class copy construct is
- a. Compiler generated default constructor
 - b. Compiler generated copy constructor
 - c. Both of these options are correct
 - d. None of these operators are correct
195. In case of protected inheritance, public members of the base class will be in derived class
- a. Private
 - b. Public
 - c. Protected
 - d. Hidden
196. Methodologies to the development of reusable software related to
- a. Structured programming
 - b. Procedural programming
 - c. Generic programming
 - d. None of the given
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- a. Its constructor initialization list
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 - c. Both from its constructor initialization list or body
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- a. Default assignment operator
 - b. User defined assignment operator
 - c. Default copy constructor

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d. None of these

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Protected:

Int I;

};

Class class2: private class1{

};

- a) Public
- b) Protected
- c) Private
- d) None of the options

200. In case of private inheritance, private members of base class will be in derived class?

- a. Private
- b. Public
- c. Protected
- d. Hidden

201. Consider the code below class class1{

Private:

Void func1();

};

Class class2: public class1{

};

Function func1 of class 1 is in class2.

- a. Public

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- b. Protected
 - c. Private
 - d. None of the given options
202. We can call base class assignment operator in derived class user defined assignment operator
- a. Implicitly
 - b. Explicitly
 - c. Using both of these options
 - d. Using none of these options
203. If the user does not specify the type of inheritance, then the default type of inheritance is
- a) Public inheritance
 - b) Private inheritance
 - c) Protected inheritance
 - d) None of given
204. Consider the code below
- ```
class class1{
public:
 void func1();
};
class class2: protected class1{
};
```
- Function func1 of class 1 is in class2.
- a. Public
  - b. Protected
  - c. Private
  - d. None of the given options

## CS304 Quiz #1

205. Which of the following causes run time binding?
- a) Declaring object of abstract class
  - b) Declaring pointer of abstract class
  - c) Declaring overridden methods as non-virtual
  - d) None of the given
206. Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?
- a) Templates (Page 256)
  - b) Overloading
  - c) Data hiding
  - d) Encapsulation
207. A copy constructor is invoked when
- a) a function do not returns by value.
  - b) an argument is passed by value. (Page 78)
  - c) a function returns by reference.
  - d) an argument is passed by reference.
208. A class template may inherit from another class template.
- a) True (Page 288)
  - b) False
209. By default, the vector data items are initialized to
- a) 0 google
  - b) 0.0
  - c) 1
  - d) Null
210. In Private ---- only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object
- a) specialization
  - b) inheritance (Page 216)
  - c) abstraction

## CS304 Quiz #1

d) composition

211. Which of the following is/are advantage[s] of generic programming?

- a) Reusability
- b) Writability
- c) Maintainability
- d) All of given (Page 256)

212. Template functions use \_\_\_ than ordinary functions.

- a) Greater Memory
- b) Lesser Memory
- c) Equal Memory
- d) None of the given options

213. Non-Template Friend functions of a class are friends of instance/s of that class.

- a) All google
- b) One specific
- c) All instances of one date type
- d) None of the given options

214. A pointer to a base class can point to objects of a derived class.

- a) True google
- b) False

215. A template argument is preceded by the keyword .

- a) vector
- b) class
- c) template
- d) type\*

216. Which one of the following terms must relate to polymorphism?

- a) Static allocation

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## CS304 Quiz #1

- b) Static typing
- c) Dynamic binding (Page 239)
- d) Dynamic allocation

217. Multiple inheritance can be of type

- a) Public
- b) Private
- c) Protected
- d) All of the given

218. Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access

- a) public members of Derv
- b) protected members of Derv.
- c) private members of Derv.
- d) protected members of Base.

219. A function call is resolved at run-time in

- a) non-virtual member function.
- b) virtual member function. (Page 239)
- c) Both non-virtual member and virtual member function.
- d) None of given

220. Two important STL associative containers are and .

- a) set, map
- b) sequence, mapping
- c) set, multiset
- d) set, map

221. An abstract class is useful when,

- a) We do not derive any class from it.
- b) There are multiple paths from one derived class to another.
- c) We do not want to instantiate its object
- d) You want to defer the declaration of the class.

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## CS304 Quiz #1

66. Suppose you create an uninitialized vector as follows:

`vector<int> evec;` After adding the statement, `evec.push_back(21);`

what will happen?

- a) The following statement will add an element to the start (the back) of `evec` and will initialize it with the value 21.
- b) The following statement will add an element to the center of `evec` and will reinitialize it with the value 21.
- c) The following statement will delete an element to the end (the back) of `evec` and will reinitialize it with the value 21.
- d) The following statement will add an element to the end (the back) of `evec` and initialize it with the value 21.

222. Default constructor is such constructor which either has no -or if it has some parameters these have -----

- a) Parameter, temporary
- b) Null, Parameter
- c) Parameter, default (Page 75)
- d) None of the given

223. Classes like `TwoDimensionalShape` and `ThreeDimensionalShape` would normally be concrete, while classes like `Sphere` and `Cube` would normally be abstract.

- a) True
- b) False

224. In order to define a class template, the first line of definition must be:

- a) `template <typename T>` (Page 257)
- b) `typename <template T>`
- c) `Template Class <ClassName>`
- d) `Class <Template T>`

225. Identify the correct way of declaring an object of user defined template class `A` for `char` type members?

- a) `A< char > obj;`

## CS304 Quiz #1

- b) <char>A obj;
- c) A obj<char>;
- d) Obj <char> A;

226. The user must define the operation of the copy constructor.

- a) True
- b) False

227. The find() algorithm

- a) finds matching sequences of elements in two containers.
- b) finds a container that matches a specified container.
- c) takes iterators as its first two arguments. (Object-Oriented Programming in C++)
- d) takes container elements as its first two arguments.

228. Compiler performs type checking to diagnose type errors,

- a) Static (Page 261)
- b) Dynamic
- c) Bound
- d) Unbound

229. Vectors contain contiguous elements stored as a[an] .

- a) variable
- b) array (Page 306)
- c) function
- d) datatype

230. In a de-queue, (chosed the best option)

- a) data can be quickly inserted or deleted at any arbitrary location.
- b) data can be inserted or deleted at any arbitrary location, but the process is relatively slow.
- c) data cannot be quickly inserted or deleted at either end.
- d) data can be inserted or deleted at either end, but the process is relatively slow

## CS304 Quiz #1

231. When you run this program, it will print out a single line to the console output  
What will be in that line?

Select one correct answer from the following list:

- a) Generic element (not sure)
- b) Heading element
- c) Paragraph element
- d) Nothing will be printed.

232. Suppose we have two derived classes from a single class, can we write a method with same name in both these derived classes?

Choose the best option.

- a) No
- b) Only if the two classes have the same name (Page 204)
- c) Only if the main program does not declare both kinds

233. When a virtual function is called by referencing a specific object by name and using the dot member selection operator (e.g., `squareObject.draw()`), the reference is resolved at compile time.

- a) True
- b) False

234. Considering the resolution order in which considering the resolution order in which compiler search for functions in a program; the first priority is given to; the first priority is given to,

- a) general template
- b) partial specialization
- c) complete specialization
- d) ordinary function (Page 287)

235. One purpose of an iterator in the STL is to connect algorithms and containers.

- a) True
- b) False

## CS304 Quiz #1

236. In , a base class can be replaced by its derived class,

- a. Sub-typing (Page 31)
- b. Super-typing
- c. Multiple-typing
- d. Restricted-typing

237. What is a class?

- a) A class is a section of computer memory containing objects.
- b) A class is a section of the hard disk reserved for object-oriented programs
- c) A class is the part of an object that contains the variables.
- d) A class is a description of a kind of object.

238. Inheritance is a way to

- a) organize data.
- b) pass arguments to objects of classes.
- c) add features to existing classes without rewriting them. (Page 27)
- d) improve data-hiding and encapsulation.

239. We can use "this" pointer in the constructor in the body and even in the initialization list of any class if we are careful,

- a. True
- b. False

240. ----- and ----- methods may not be declared abstract.

- a) private,static
- b) private,public
- c) static,public
- d) none of given

241. In \_\_\_\_\_, a base class can be replaced by its derived class,

- a) Sub-Typing
- b) Super-Typing
- c) Multiple-Typing
- d) Restricted Typing

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## CS304 Quiz #1

242. Public methods of base class can -----be accessed in its derived class

- a) directly (Page 179) rep
- b) indirectly
- c) simultaneously
- d) None of the given

243. What is true about function templates?

- a) The compiler generates only one copy of the function template
- b) The compiler generates a copy of function respective to each type of data (Page 256)
- c) The compiler can only generate copy for the int type data
- d) None of the given.

244. When the base class and the derived class have a member function with the same name, you must be more specific which function you want to call (using \_ ).

- a) scope resolution operator
- b) dot operator
- c) null operator
- d) Operator overloading

245. one this pointer does not point to current object of any class,

- a) True
- b) False

246. Which of the following operator(s) take(s) one or no argument if overloaded?

- a) ++(Page 162)
- b) -
- c) +
- d) All of the given

247. Which of the following operators cannot be overloaded?

- a) Scope resolution operator ( :: ) (Page 141)
- b) Insertion operator ( << )

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- c) Extraction operator ( >> )
- d) The relation operator ( > )

248. Which of these are examples of error handling techniques?

- a) Abnormal Termination
- b) Graceful Termination
- c) Return the illegal
- d) all of the given (Page 329)

249. Virtual functions allow you to

- a) Create an array of type pointer-to-base class that can hold pointers to derived classes.
- b) Create functions that can never be accessed.
- c) Group objects of different classes so they can all be accessed by the same function code.
- d) use the same function call to execute member functions of objects from different classes

250. The copy () algorithm returns an iterator to

- a) the last element copied from.
- b) the last element copied to.
- c) the element one past the last element copied from.
- d) the element one past the last element copied to.

251. If you define a vector v with the default constructor, and define another vector w with a one-argument constructor to a size of 11, and insert 3 elements into each of these vectors with push\_back(), then the size() member function will return for v and for w.

- a) 11 for v and 3 for w.
- b) 0 for v and 0 for w.
- c) 0 for v and 3 for w.
- d) 3 for v and 11 for w.

252. 97. Which is not the Advantage of inheritance?

- a) providing class growth through natural selection. (Object-Oriented Programming in C++)
- b) facilitating class libraries.
- c) avoiding the rewriting of code.
- d) providing a useful conceptual framework.

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253. How the information hidden within an object can be accessed?

- a) Through its interface
- b) Through its private data members
- c) Through its private member functions
- d) Through both public and private members

254. Encapsulation means Select correct option:

- a) Extending the behavior of class in another class
- b) Data and behavior are tightly coupled within an entity (Page 16)
- c) One entity takes all the attributes and operations of the other
- d) Taking out the common features and put those in a separate class

255. An STL container cannot be used to,

- a) hold objects of class employee.
- b) store elements in a way that makes them quickly accessible.
- c) compile c++ programs. (Object-Oriented Programming in C++)
- d) organize the way objects are stored in memory

256. Which of the following may not be an integral part of an object?

- a) State
- b) Behavior
- c) Protected data members (Page 12)
- d) All of given

257. Suppose that the Test class does not have an overloaded assignment operator.

What happens when an assignment  $a=b$ ; is given for two Test objects a and b?

- a. The automatic assignment operator is used
- b. The copy constructor is used
- c. Compiler error
- d. Run-time error

## CS304 Quiz #1

258. A static member function can be called, even when a class is not \_\_\_\_\_.  
a. Declared  
b. Define  
c. Instantiated  
d. Called
259. What problem(s) may occur when we copy objects without using deep copy constructor?  
a. Dangling pointer  
b. Memory Leakage  
c. All of the given  
d. System crash
260. \_\_\_\_\_ provide the facility to access the data member.  
a) accessor function  
b) private function  
c) inline function  
d) None of the given
261. The \_\_\_\_\_ relationship indicates that an object contains other objects.  
a. None of given  
b. 'has-a'  
c. 'is-a'  
d. 'be'
262. Which one of the following features of OOP is used to derive a class from another?  
a. Encapsulation  
b. Polymorphism  
c. Data hiding  
d. Inheritance
263. \_\_\_\_\_ satisfy the condition of polymorphism  
a. Carbon



## CS304 Quiz #1

- b. Diamond
  - c. Coal
  - d. all of the given
264. A generalization-specialization relation between classes are implemented using
- a) data hiding
  - b) friend classes
  - c) encapsulation
  - d) inheritance
265. Which of the following is not an example of multiple inheritances?
- a) Mermaid
  - b) Woman
  - c) None of the given
  - d) Amphibious Vehicle
266. A generic class showing all the common attributes and a behavior of other classes represents a very important feature in oop called ---
- a) Inheritance
  - b) Encapsulation
  - c) Abstraction
  - d) Polymorphism
267. The parameters given in temple definition other than those used for mentioning templates types are called
- a) Type Parameters
  - b) Non Type parameters
  - c) Default Type Parameters
  - d) None of the given option
268. Which of the following is the correct way to define a template class X?
- a) Class < typename T>class X{}
  - b) Template class C{}
  - c) Template < typename T> classX{}

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- d) `typename<class T > class X{}`
269. In resolution order of function template, compiler searches for in the end
- a) Complete specialization
  - b) Partial specialization
  - c) Ordinary function
  - d) Generic template
270. may inherit from a complete specialization
- a. Partial specialization
  - b. Complete specialization
  - c. Ordinary class
  - d. All of given options
271. Which of the following may inherit from the ordinary class?
- a. Class template
  - b. Partial specialization
  - c. Complete specialization
  - d. All of given option
272. Which of the following is a sequence container?
- a. Map
  - b. Vector
  - c. Set
  - d. multiset
273. Each \_of a template class by default becomes function template
- a. Data member
  - b. Member function
  - c. Type parameter
  - d. Object

## CS304 Quiz #1

274. When we specialize a function template, it is called
- a. Function overloading
  - b. Function overriding
  - c. Function template overloading
  - d. Function template overriding
275. Which of the following is NOT an associative container?
- a. Set
  - b. Multiset
  - c. Map
  - d. Group
276. A is a pointer which is declared outside the
- a. Container, cursor
  - b. Cursor, container
  - c. Container, class
  - d. This, cursor
277. Which of the following are collectively referred to as the first class containers?
- a. Associative containers and containers adapters
  - b. Sequence containers and associative containers
  - c. Sequence containers and container adapters
  - d. None of the given options
278. Which of the following can be passed as type argument to template?
- a) Primitive type
  - b) User defined types
  - c) Both primitive type and user defined types
  - d) None of the given options
279. Which of the following is known as Dereference operator inC++?
- a) +

# CS304 Quiz #1

b) \*

c) &

d) d :: :

280. Which of the following represents partial specialization?

a. Template < int, char >

b. Template < class T, Type t>

c. Template<class T, Class U, int>

d. Template< typename T, class W>

281. When we declare template function as friend of any class, then which of the following specialization(s) of template function also become friend of the class granting friendship?

a) Explicit

b) Implicit

c) Partial

d) All of the given options

282. Which of the following represents complete specialization?

a) Template < int, char >

b) Template < class T, float>

c) Template<class T, Class U , int>

d) Template< class , class W>

283. Which of the following is/are the key component of standard template library?

a) Containers

b) Iterators

c) Algorithm

d) All of the given option

284. Which of the following statement is true about partial specialization?

a) Function templates cannot have partial specialization

b) Class template cannot have partial specialization

c) Both class templates and function template can have partial specialization.

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d) None of the given

285. Which of the following is correct code portion to initialize static datamember "value" of int type for class X?

- a) `Template <class T> X::value =0;`
- b) `Template<class T>int X::value =0;`
- c) `Template<class T> int X < T >::value = 0;`
- d) `Template<class T> X < T >::value =0;`

286. When we want to have exactly identical operations on different data type, are \_\_\_\_ used

- a) Function Template
- b) Function Overloading
- c) Function Overriding
- d) None of the given option

287. In order to define a class template the first line of definition must be:

- a) `Template <typename T>`
- b) `typename < template T>`
- c) `Template class <ClassName>`
- d) `Class <Template T>`

288. In case of template specialization, if compiler cannot find required complete specialization then it searches for some

- a) General template
- b) Complete template
- c) Partial specialization
- d) None of the Given

289. Which of the following is a constrained version of some first-class container?

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## CS304 Quiz #1

- a) Sequence containers
- b) Associative containers
- c) Container Adapter
- d) None of the given options

290. In resolution order, highest priority is given to in template specialization.

- a) General template
- b) Complete specialization
- c) Partial specialization
- d) None of the given options

291. Which of the following is not a sequence container?

- a) Vector
- b) Deque
- c) Map
- d) List

292. STL stands for:

- a) Secure Template Library
- b) Structured Template Library
- c) Stack Template Library
- d) Standard Template Library

293. \_\_\_\_\_ Binding means that target function for a call is selected at compile time.

- a) Static
- b) Dynamic
- c) Automatic
- d) None of the given

294. In C++, Composition is a relationship between \_\_\_\_\_ and \_\_\_\_\_ objects.

- a) Parent, child

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- b) Base, derived
- c) Whole, part
- d) All of the given

295. A pure virtual function is a virtual function that

- a) causes its class to be abstract.
- b) returns nothing.
- c) Is used in a derived class.
- d) takes no arguments.

296. If class B inherits from class A then it contains all characteristics of \_\_\_\_\_.

- a) Exception Handling
- b) Inheritance
- c) Polymorphism
- d) Information Hiding

297. Polymorphism makes the system:

- a) reusable
- b) flexible
- c) faster
- d) All of the given

298. \_\_\_\_\_ is represented by a line with an unfilled diamond head towards the container.

- a) Inheritance
- b) Association
- c) Aggregation
- d) Composition

299. Subtyping means that derived class is behaviorally \_\_\_\_\_ with the base class.

- a) Same
- b) Compatible
- c) Different
- d) Incompatible

## CS304 Quiz #1

300. In operator overloading, which of the following operator takes one or no argument.

- a) ++
- b) \*
- c) %
- d) <<

301. In C++ generic programming is done using 148. 149. 150. 151. 152. 153.

- a) Procedures
- b) Packages
- c) Templates
- d) None of the given

302. Function overriding is done in context of,

- a) Single class
- b) Single derived class
- c) Single base class
- d) Derived and base class

303. Which of the following can be the behavior of an object "Usman"?

- a) Name
- b) Age
- c) Address
- d) Eat

304. In polymorphism, messages can be interpreted in different ways depending upon the \_\_\_\_\_ class.

- a) base
- b) derived
- c) sender
- d) receiver



## CS304 Quiz #1

305. Information hiding can be achieved through\_\_\_\_\_.

- a) Encapsulation, Inheritance
- b) Encapsulation, Polymorphism
- c) Encapsulation, Abstraction
- d) Overloading

306. A good model is ..... related to a real life problem.

- a) Loosely
- b) Openly
- c) Closely

307. Which of the following features of OOP is used to derive a class from another?

- a) Encapsulation
- b) Polymorphism
- c) Data hiding
- d) Inheritance

308. Which of the following is a weak relationship between two objects?

- a) Inheritance
- b) Composition
- c) Aggregation
- d) None of given

309. Data items in a class must be private.

- a) True
- b) False

310. Which one is a class association?

- a) Simple Association
- b) Inheritance

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- c) Composition
- d) Aggregation

311. Suppose there is an object of type Person, which of the following can be considered as one of its attributes

- a) Name
- b) Age
- c) Work()
- d) Both Name and Age

312. Which one is not an object association?

1. Simple association

2. Inheritance

3. Aggregation

4. Association

313. Using encapsulation we can achieve

- a) Information hiding
- b) Least interdependencies among modules
- c) Implementation independence
- d) All of given options

314. In constant member function the type of this pointer is:

- a) Constant pointer
- b) Constant pointer to object
- c) Constant pointer to class

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## CS304 Quiz #1

d) Constant pointer to constant object

315. Which of the following is the way to extract common behavior and attributes from the given classes and make a separate class of those common behaviors and attributes?

- a) Generalization
- b) Sub-typing
- c) Specialization
- d) Extension

316. The ability to derive a class from more than one class is called

- a) Single inheritance
- b) Encapsulation
- c) Multiple inheritance
- d) Polymorphism

317. If MyClass has a destructor what is the destructor named?

- a) MyClass
- b) ~MyClass
- c) My~Class
- d) MyClass~

318. Class abc{ ----- }; Is a valid class declaration?

- a) yes
- b) no

319. Without using Deep copy constructor, A \_\_\_\_\_ problem can occur

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- a) System crash
- b) Memory Leakage
- c) Dangling pointer
- d) All of the given

320. If only one behavior of a derived class is incompatible with base class, then it is:

- a) Generalization
- b) Specialization
- c) Extension
- d) Inheritance

321. Which of the following may not be an integral part of an object?

- a) state
- b) behavior
- c) Protected data members
- d) All of given

322. Only tangible things can be chosen as an object.

- a) True
- b) False

323. Class is not a mechanism to create objects and define user data types.

- a) true
- b) false

324. Memory is allocated to non-static members only, when:

- a) Class is created
- b) Object is defined
- c) Object is initialized
- d) Object is created

## CS304 Quiz #1

325. The sub-object's life is not dependent on the life of master class in \_\_\_\_.

- a) Composition
- b) Aggregation
- c) Separation
- d) none of the given

326. Unary operators and assignment operator are right associative.

- a) true
- b) false

327. The  $\geq$  operator can't be overloaded.

- a) true
- b) false

328. \_\_\_\_ is creating objects of one class inside another class.

- a) Association
- b) Composition
- c) Aggregation
- d) Inheritance

329. If we are creating array of objects through new operator, then

- a) We can call overloaded constructor through new
- b) We can't call overloaded constructor through new
- c) We can call default constructor through new
- d) None of the given

330. Object can be declared constant with the use of Constant keyword.

- a) true
- b) false

331. \_\_\_\_\_ Operator will take only one operand.

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## CS304 Quiz #1

- a) New
- b) int
- c) object
- d) none of the given

332. Which of the following operator(s) take(s) one or no argument if overloaded?

- a) ++
- b) \*
- c) %
- d) All of the given choices

333. This pointer does not pass implicitly to \_\_\_\_\_ functions.

- a) Static Member
- b) Non-Static Member
- c) Instance Number
- d) None of the given

334. Operator overloading is

- a) making C++ operators work with objects.
- b) giving C++ operators more than they can handle.
- c) giving new meanings to existing Class members.
- d) making new C++ operators

335. Which of the following is TRUE,

- a) Derived class pointer can be used as Base class pointer
- b) Base class pointer can be used as Derived class pointer
- c) Both of these options
- d) None of these options

336. \_\_\_\_\_ Binding means that target function for a call is selected at run time

- a) Automatic
- b) Dynamic

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- c) Static
- d) Dramatic

337. When we want to implement one class in terms of another class then we use

- a) Public inheritance
- b) Protected inheritance
- c) Private inheritance
- d) None of these options

338. Consider the code below, class c1{ }; class c2 : public c1 { }; class c3 :

public c2 { }; Then c2 is

- a) Direct base class of c3
- b) Direct child class of c3
- c) Direct base class of c1
- d) None of these

A parent class can call constructor of its child class through

- a) Its constructor initialization list
- b) Its constructor body
- c) Both from its constructor initialization list or body
- d) Can not call the constructor of its child class

339. In Private Inheritance the public members of base class become \_\_\_\_\_ in derived class.

- a. Public
- b. Private
- c. Protected
- d. None of the given options

## CS304 Quiz #1

340. Function overriding is done in context of,
- a) Single class
  - b) Single derived class
  - c) Single base class
  - d) Derived and base classes
341. Child class can call constructor of its,
- a) Direct base class
  - b) Indirect base class
  - c) Both direct and indirect base classes
  - d) None of these
- a) Adding a derived class to a base class requires fundamental changes to the base class
- a. True
  - b. False
342. \_\_\_\_\_ is automatically called when the object is created.
- a. Member function
  - b. Object
  - c. Constructor
  - d. None of the given
343. Inheritance is a way to
- a. Organize data
  - b. Pass arguments to objects of classes
  - c. Add features to existing classes without rewriting them
  - d. Improved data-hiding and encapsulation



## CS304 Quiz #1

344. Through interface we access object\_\_\_\_\_.
- a. States
  - b. Data members
  - c. Behaviour
  - d. None of the given
345. Which of the following may not be integral part of an object?
- a) State
  - b) Behavior
  - c) Protected data members
  - d) All of given
346. If we have an overloaded constructor in our class, then compiler construct the default constructor.
- a. True
  - b. False
347. Advantages(s) of information hiding
- a. Simplifies the model
  - b. Restriction to change
  - c. Both of above
  - d. None of the above
348. If some of objects exhibit identical characteristics, then they belong to:
- a. Different classes
  - b. Multiple classes
  - c. Same class

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## CS304 Quiz #1

- d. None of the given
349. Which of the following is the way to extract common behaviour and attributes from the given and make a separate class of those common behaviours and attributes?
- a. Generalization
  - b. Sub-typing
  - c. Specification
  - d. Extension
350. A class has \_\_\_\_\_ destructor
- a. One
  - b. Two
  - c. Three
  - d. Four
351. A class has \_\_\_\_\_ constructor
- a. One
  - b. Two
  - c. Three
  - d. Four
352. Behaviors of a type of objects are represented through
- a. States
  - b. Attributes
  - c. Operations
353. Which is not an example of multiple inheritances?
- a. Mermaid
  - b. Amphibious Vehicle

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c. Car

d. None of the above

354. Static data members are called \_\_\_\_\_ variable

a. Class

b. Object

c. Structure

d. None of the given

355. Associativity can be change in operator overloading.

a. True

b. False

356. \_\_\_\_\_ and \_\_\_\_\_ methods may not be declared abstract.

a. Private,static

b. private,public

c. static,public

d. none of the given

357. Let Suppose a class Student with objects std1, std2, and std3. For the statement std3 =

std1 - std2 to work correctly, if the overloaded - operator must

a. take two arguments.

b. None of the given choices

c. take single argument

d. take three arguments

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# CS304 Quiz #1

358. To initialize an array of objects, only \_\_\_\_\_ will be called
- a. Default Constructor
  - b. Overloaded Constructor
  - c. Default Object
  - d. None of the above
359. \_\_\_\_\_ provide the facility to access the data member.
- a. accessor function
  - b. private function
  - c. inline function
  - d. None of the given
360. If a class involves dynamic memory allocation, then:
- a. Default copy constructor, shallow copy is implemented
  - b. User defined copy constructor, shallow copy is implemented
  - c. Default copy constructor, deep copy is implemented
  - d. User defined copy constructor, deep copy is implemented
361. The sentence "Object Oriented Programming book in bookshelf" is an example of:
- a. Association
  - b. Multiple associations
  - c. Aggregation
362. How the information hidden within an object can be accessed?
- a. Through its interface
  - b. Through its private data member

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- c. Through its private member functions
363. There is no need to is or is a kind of relationship in generalization
- a. True
  - b. False
364. An object has \_\_\_\_\_ interface(s)
- a. Only one
  - b. Only two
  - c. One or more than one
365. Which of the following is an advantage of OOP?
- a. It provides an ability to create one user defined data type extending the other
  - b. It provides the facility of defining abstract data type through which real entities can be defined better
  - c. All of the given option
366. Encapsulation means
- a. Extending the behaviour of class in another class
  - b. Data and behaviour are tightly coupled within an entity
  - c. One entity takes all the attributes and operations of the other
  - d. Taking out the common features and put those in a separate class
367. We can allocate dynamic memory allocation through \_\_\_\_\_ key word.
- a. Static
  - b. Global
  - c. New

## CS304 Quiz #1

- d. Const
368. In programming, where the actual actions are coded is called:
- a. Function declaration
  - b. Function calling
  - c. Function definition
  - d. None of the given
369. Which of the following is directly related to polymorphism? :
- a. Overriding
  - b. Const members
  - c. Static members
  - d. None of given
370. In a class declaration, data or functions designated private are accessible
- a. to any function in the program.
  - b. only if you know the password.
  - c. to member functions of that class.
  - d. 4. only to public members of the class.
371. An instance of user defined type is called
- a. Object
  - b. Class
  - c. both of above
  - d. none of above
372. Aggregation is implemented using pointer.
- a. True
  - b. False

## CS304 Quiz #1

373. If class A supports all the operations of class B, then class A is:

- a. Class A is behaviourally compatible with class B
- b. Class A is behaviourally incompatible with class B
- c. Class A is independent of class B
- d. None of the given

374. Objects having identical characteristics belong to \_\_\_\_\_

- a. Same class
- b. Two different classes
- c. Any number of different classes
- d. Objects can not have identical characteristics



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## CS304 GRAND QUIZ

| SR NO | MCQS QUESTION                                                                                                                   | ANSWER                                                  |
|-------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 1     | In expression $c1 * c2 + c3 - c4$ which of the following will be executed in first order                                        | $C1 * c2$                                               |
| 2     | Inheritance is a way to                                                                                                         | Add features to existing classes without rewriting them |
| 3     | ----- represent "IS A" relationship                                                                                             | Inheritance                                             |
| 4     | Constructor without any parameter is called:                                                                                    | Default constructor                                     |
| 5     | Which of the following stores the address of a variable?                                                                        | Pointer                                                 |
| 6     | A Post-fix unary operator is implemented in C++ using Member function with                                                      | 1 dummy int argument                                    |
| 7     | minimum classes required in a program for implementing multiple inheritance                                                     | 3                                                       |
| 8     | Which of the following statement(s) describe constant member functions?                                                         | All of the given option                                 |
| 9     | composition is a strong relation relationship because                                                                           | Composed object cant exist independency                 |
| 10    | How many interactions are there in the following scenario Imran has the car. he drive the car and car has four wheels and doors | 3                                                       |
| 11    | if we extend our model ,and the rest of the model is not affected then it is called-----                                        | Flexibility                                             |
| 12    | choose correct declaration of overload $*=1$ operator for class complex as Member function                                      | Complex and operator $*=(const complex \& c)$           |
| 13    | Consider the following class <code>A{int a,b,c;</code>                                                                          | A,b,c                                                   |

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|    |                                                                                                                                     |                                            |
|----|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
|    | public: A():B(10),C(0)a(7) };                                                                                                       |                                            |
| 14 | Class is a blue print of                                                                                                            | Objects                                    |
| 15 | Suppose we have a class student which of the following statement is correctly declaring the pointer to the object of student class? | Student* object                            |
| 16 | In c++ composition is relationship between -----and-----objects                                                                     | Whole, part                                |
| 17 | Which of the following Depicts the proper definition of class?                                                                      | a class is description of a kind of object |
| 18 | in class attributes are represented by ---- and----- respectively                                                                   | data member, Member function               |
| 19 | which of the following represents the two Association                                                                               | Employee works for the company             |
| 20 | Entities in OO model are represented by                                                                                             | Rectangle                                  |
| 21 | Which of the following features of OOP is used to derive a class from another?                                                      | Inheritance                                |
| 22 | In C ++ which of the following is defined as a stream insertion operator?                                                           | <<                                         |
| 23 | choose the correct declaration of overloaded equality (==)operator for class string as a Member function                            | Bool operator==(const string and s)        |
| 24 | ---- is the relationship between the part object and the whole object                                                               | Composition                                |
| 25 | object hides its information from others this concept is known as                                                                   | information hiding                         |
| 26 | the concept of derived classes is involved in                                                                                       | Inheritance                                |
| 27 | abstraction provides information according to                                                                                       | user perspective                           |
| 28 | which of the following is not an object Association                                                                                 | Inheritance                                |
| 29 | which of the following classes are used by amphibious vehicle to inherit                                                            | both land and water vehicle                |

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|    |                                                                                                             |                                                               |
|----|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
|    | characteristics                                                                                             |                                                               |
| 29 | object data includes                                                                                        | both states and attributes of an object                       |
| 30 | choose correct declaration of overloaded inequality operator for class string as non member friend function | friend pool operator !<br>=(const string &s1,const string &2) |
| 31 | How many objects of a given class may be constructed in an application?                                     | as many as the application ask for                            |
| 32 | how many objects of a given class may be constructed in an application                                      | as many as the application ask for                            |
| 33 | Which of the following statements describe constant member functions?                                       | all of the given option                                       |
| 34 | student *ptr=new student(" Ali"); is an example of                                                          | static memory allocation                                      |
| 35 | in----- base class can be replaced by the derived class                                                     | extension                                                     |
| 36 | Which of the following is a correct way of declaring constant variable ROLLNO?                              | cosnst int Roll no;                                           |
| 37 | which of the following operator doesn't take argument hit overloaded                                        | none of the given choices                                     |
| 38 | which of the following is part of the Class diagram                                                         | all of the given option                                       |
| 39 | a child inherits characteristic from its                                                                    | parent                                                        |
| 40 | which of the following issues can be caused by the use of friend function                                   | programming Bugs                                              |
| 41 | which of the following is not true about constant number function                                           | all of the given option                                       |
| 42 | Which one is not the main features of object oriented programming?                                          | exception handling                                            |
| 43 |                                                                                                             |                                                               |
| 44 | in object oriented programming, objects communicate with each other through                                 | messages                                                      |
| 45 | static data member is declared                                                                              | inside the class                                              |
| 46 | the concept of derived classes is involved                                                                  | inheritance                                                   |

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|    |                                                                                                                                               |                                             |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
|    | in                                                                                                                                            |                                             |
| 47 | Which of the following can we attribute of an object "Ali"?                                                                                   | address                                     |
| 48 | Which of the following keyword tell the compiler to substitute the code within the function definition for every instance of a function call? | inline                                      |
| 49 | inheritance is the relationship between                                                                                                       | classes                                     |
| 50 | how many objects are involved in the binary Association                                                                                       | 2                                           |
| 51 | all objects of same class has ----- attributes                                                                                                | same                                        |
| 52 | which of the following concept is more close to encapsulation                                                                                 | information hiding                          |
| 53 | member initialization list is used to                                                                                                         | initialize data member                      |
| 54 | aggregation between objects show that                                                                                                         | objects have weak relationship              |
| 55 | there is /are ways to create user defined types for objects in C ++                                                                           | two                                         |
| 56 | composition is a strong relationship because:                                                                                                 | composed object can not exist independently |
| 57 | in polymorphism messenger can be interpreted in different ways depending upon the----- class                                                  | receiver                                    |
| 58 | which of the following is most suitable advantage of information hiding                                                                       | simplifies the model                        |
| 59 | the process of extracting common characteristics from two or More classes and combining them into a generalized superclass is called          | Generalization                              |
| 60 | ----- operator is used to create objects at run time                                                                                          | new                                         |
| 61 | choose the correct option to declare an overloaded Stream insertion operator for                                                              | friend ostream and operator<<(ostream       |

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|    |                                                                                                                    |                               |
|----|--------------------------------------------------------------------------------------------------------------------|-------------------------------|
|    | userdefined class "string" as a non-member friend function                                                         | and os,const string & s)      |
| 62 | static data member is declared                                                                                     | inside the class              |
| 63 | an instance of user defined type is called                                                                         | object                        |
| 64 | which of the following is true about constructor                                                                   | all of the given option       |
| 65 | Which of the following correct way of declaring constant variable ROLLNO?                                          | const int ROLLNO;             |
| 66 | inheritance is shown by                                                                                            | filled arrow                  |
| 67 | which of the following operators operate on one operand                                                            | unary operator                |
| 68 | which of the following will happen when a constant function tried to change the value of data members of the class | compile time error will occur |
| 69 | in object oriented programming, objects communicate with each other through---                                     | messages                      |
| 70 | How the information hidden within an object can be accessed?                                                       | through its interface         |
| 71 | object orientation approach can be achieved using object and their-----                                            | interactions                  |
| 72 | constructors have ----- return type                                                                                | no                            |
| 73 | which of the following is an example of unary operator                                                             | increment operator(++)        |
| 74 | memory is allocated to non-static members only when                                                                | object is created             |
| 75 | in class B inherit from class A which of the following is child class                                              | class B                       |
| 76 | the----- tells the compiler what task the function will be performing                                              | function definition           |
| 77 | Consider the statement" room has chair" which of the following type of Association exists between whom and chair?  | aggregation                   |
| 78 | how many objects are involved in the N-                                                                            | more than 3                   |

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|    |                                                                                                           |                                                      |
|----|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------|
|    | ary Association                                                                                           |                                                      |
| 79 | deconstruct is a function which has the same name as that of class but starts with a----- sign            | ~                                                    |
| 80 | which of the following is a tangible entity                                                               | car                                                  |
| 81 | member initializer list is used to                                                                        | initialize data members                              |
| 82 | we can get the address of variable stored in pointer using-----                                           | & symbol                                             |
| 83 | two sentence object oriented programming book in bookshelf is an example of                               | aggregation                                          |
| 84 | which of the following best defines the obstruction                                                       | hide the implementation and show current perspective |
| 85 | which of the following define default constructor 5 class student                                         | student(){//..}                                      |
| 86 | object data includes                                                                                      | both state and attributes of an object               |
| 87 | minimum classes required in a program for implementing multiple inheritance                               | 3                                                    |
| 88 | information hiding can be achieved through-----                                                           | encapsulation, abstraction                           |
| 89 | compiler generated defaulter structure is called----- hand user written default constructor is called---- | implicit, explicit                                   |
| 90 | which of the following is true about constructor                                                          | all of the given option                              |
| 91 | In which of the following different forms office single entity exist?                                     | polymorphism                                         |
| 92 | data members of a class student can be:                                                                   | all of the given option                              |
| 93 | the----- turn off the compiler what does the function will be performing                                  | function definition                                  |
| 94 | Suppose a class does not have any                                                                         | compiler will call                                   |

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|     |                                                                                                                                              |                                                                                                  |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
|     | constructor what will happen when an object of this class is created?                                                                        | implicit default constructor                                                                     |
| 95  | how can we differentiate between constructor and destructor                                                                                  | destructor are preceded with a Tilda(~) symbol, and constructor are not preceded with any symbol |
| 96  | ---- is used for hiding the details of an object from the other parts of a program                                                           | encapsulation                                                                                    |
| 97  | Class diagram include                                                                                                                        | class name, attributes, operation                                                                |
| 98  | through interface was access objects-----                                                                                                    | behavior                                                                                         |
| 99  | composition is a strong relationship because                                                                                                 | compose object cannot exist independently                                                        |
| 100 | copy constructor is called when:                                                                                                             | an object is created in term of Preexisting object                                               |
| 101 | choose the correct option to declare an overloaded stream insertion operator for a user defined class string as a non-Member friend function | friend ostream & operator(ostream &os.const string &s)                                           |
| 102 | a post-fix unary operator implemented in C plus plus using non Member function with---- arguments                                            | two                                                                                              |
| 103 | in class attributes and behaviors are represented by---- and--- respectively                                                                 | data members, member functions                                                                   |
| 104 | which of the following best describes the relationship between book and book chapter classes                                                 | composition                                                                                      |
| 105 | which of the following is true about destructor                                                                                              | all of the given option                                                                          |
| 106 | in composition----- are called from composing objects to composed subjects                                                                   | destructors                                                                                      |
| 107 | which of the following can be used an                                                                                                        | Global variables                                                                                 |

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|     |                                                                                                                  |                                            |
|-----|------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
|     | alternative to Static members                                                                                    |                                            |
| 108 | which of the following statement is true about static function of a class                                        | it is used to access static data members   |
| 109 | which of the following operator does not take any arguments if overloaded                                        | none of the given choices                  |
| 110 | what is the output of the following code<br><code>int main(){int cost x=10;cout&lt;&lt;+<br/>+x;return0;}</code> | error                                      |
| 111 | abstraction includes-----                                                                                        | relevant information                       |
| 112 | Which of the following can be the behavior of an object "Usman"?                                                 | eat                                        |
| 113 | a class defined within another class is                                                                          | nested class                               |
| 114 | when an object is initialized, is automatically called by the compiler                                           | constructor                                |
| 115 | the compiler treats the Member function of a class as                                                            | inline function                            |
| 116 | an object has attributes operation ,and-----                                                                     | unique identity                            |
| 117 | Which of the following is not an access specifier in C ++?                                                       | hidden                                     |
| 118 | which of the following parts of an object exhibits its state                                                     | data                                       |
| 119 | ---- is the relationship between the part object and the whole object                                            | composition                                |
| 120 | how can we identify classes from a given problem statement                                                       | by extracting nouns from problem statement |
| 121 | what "friend" should be placed in                                                                                | function declaration                       |
| 122 | generalization is----- approach                                                                                  | bottom-up                                  |
| 123 | static data members can be accessed through-----                                                                 | dot operator and scope resolution operator |
| 124 | composition age----- and aggregation is----- relationship                                                        | strong, weak                               |
| 125 | suppose there is an object of type                                                                               | name                                       |



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|     |                                                                                                                                                                |                                                               |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
|     | person, which of the following can be considered as one of its attributes                                                                                      |                                                               |
| 126 | Which of the following depicts the proper definition of class?                                                                                                 | a class is a description of a kind of object                  |
| 127 | a child inherits characteristics from its-----                                                                                                                 | parent                                                        |
| 128 | the car is composed of wheels what is the relationship between car and wheels?                                                                                 | composition                                                   |
| 129 | ----- is creating objects of one class inside another class                                                                                                    | composition                                                   |
| 130 | the other name of specialization is-----                                                                                                                       | restriction                                                   |
| 131 | suppose a is an int type static data member class test then which of the following is correct way of initialization the static variable outside the class test | int test::a=0                                                 |
| 132 | Which of the following stores the address of variable?                                                                                                         | Pointer                                                       |
| 133 |                                                                                                                                                                |                                                               |
| 134 |                                                                                                                                                                |                                                               |
| 135 | static data member is declared                                                                                                                                 | inside the class                                              |
| 136 | polymorphism makes the system                                                                                                                                  | all of the given option                                       |
| 137 | which of the following statement best describes the constructor                                                                                                | constructor is used to initialize the data members of a class |
| 138 | data member(s) of a class student can be                                                                                                                       | all of the given option                                       |
| 139 | which of the following is a strong relationship                                                                                                                | composition                                                   |
| 140 | all objects of same class has----- attributes                                                                                                                  | same                                                          |
| 141 | in C ++ composition is a relationship between----and---- objects                                                                                               | whole, part                                                   |
| 142 | suppose 12 class string assignment operator is overloaded with following                                                                                       | compiler will generate compile time error                     |

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|     |                                                                                                                                                                                |                                                           |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
|     | declaration void operator=(const string &); what will happen when we will write following statement in main()<br>str=str2=str3;where,str1.str2,str3 are object of class string |                                                           |
| 143 | in class B inherits from class A then it contains all characteristics of----                                                                                                   | class A                                                   |
| 144 | which of the following is not an example of multiple inheritance                                                                                                               | car                                                       |
| 145 | which of the following can be created without any object of a class?                                                                                                           | static data members                                       |
| 146 | consider the following code segment<br>class test{int a;int b;int c;}                                                                                                          | junk value,5, junk value                                  |
| 147 | object orientation approach can be achieved using objects and their-----                                                                                                       | interactions                                              |
| 148 | member functions defined inside a class declaration are----- by default                                                                                                        | private                                                   |
| 149 | how can we identify classes from a given problem statement                                                                                                                     | by extracting nouns from problem statement                |
| 150 | which of the following statement is not true about static variable of a class?                                                                                                 | static variable belongs to particular instance of a class |
| 151 | there are----- types of constructor in C++                                                                                                                                     | 3                                                         |
| 152 | statement yasir is a friend of Ali and Ali is a friend of yasir is an example of                                                                                               | two way Association                                       |
| 153 | this pointer cannot be passed implicitly to----- functions                                                                                                                     | static member                                             |
| 154 | which of the following issues can be caused by the use of friend function                                                                                                      | all of the given                                          |
| 155 | in----- send a message does not need to know exact class of the receiver                                                                                                       | none of the given option                                  |

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|     |                                                                                                                         |                                           |
|-----|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| 156 | minimum classes required in a program for implementing multiple inheritance                                             | 3                                         |
| 157 | -----Associates the object of the exact two classes                                                                     | binary Association                        |
| 158 | main characteristics of information hiding is                                                                           | keep the model simple                     |
| 159 | static data member is declared                                                                                          | inside the class                          |
| 160 | in expression $b=c/d$ , the arity of division(/) operator is                                                            | 2                                         |
| 161 | we can allocate dynamic memory in C ++ through operator                                                                 | new                                       |
| 162 | Which of the following is true about accessor functions?                                                                | all of the given option                   |
| 163 | in---- base class cannot always be replaced by the derived class                                                        | specialization                            |
| 164 | abstraction help to                                                                                                     | reduce the complexity                     |
| 165 | relationship in which child object gets destroyed if parent object is destroyed                                         | composition                               |
| 166 | object oriented programming is better than procedural programming because:                                              | it is more close to reality               |
| 167 | The derived class inherits behavior from the base class and also exhibits its own behavior. this concept is known as--- | Overriding                                |
| 168 | the other name of subtyping is                                                                                          | Extension                                 |
| 169 | A good model is----- related to life problem                                                                            | closely                                   |
| 170 | in generalization general class represents the ----- behavior of the class                                              | common                                    |
| 171 | how can we identify from a given problem statement?                                                                     | by extracting noun from problem statement |
| 172 | which of the following is not a access specifier in C ++                                                                | hidden                                    |
| 173 | in composition----- are called from composing objects to composed objects                                               | destructors                               |

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|     |                                                                                                                                                                                |                                                           |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| 174 | a class have only---- destructor                                                                                                                                               | one                                                       |
| 175 | suppose you have following C + + statement<br>int old value=10; int new value=++old value; what will be the value of old value and newValue after executing above statements ? | oldVlue=11,newVlue=11                                     |
| 176 | in constant Member function the type of this pointer is                                                                                                                        | constant pointer to constant data                         |
| 177 | ----- is creating objects of one class inside another class                                                                                                                    | composition                                               |
| 178 | ----- Associates the objects of the exact three classes                                                                                                                        | ternary Association                                       |
| 179 | generalization is implemented through                                                                                                                                          | inheritance                                               |
| 180 | which of the following is strong relationship                                                                                                                                  | composition                                               |
| 181 | composition is a strong relationship because                                                                                                                                   | composed object cannot exist independently                |
| 182 | ----- is set of function of an object exposed of other objects                                                                                                                 | interface                                                 |
| 183 | suppose student is a class, which of the following defines constructor with one parameter for class student                                                                    | student(char* aName);                                     |
| 184 | which of the following is not true about constant Member function                                                                                                              | all of the given option                                   |
| 185 | ----- define the order of evaluation of an operator is an expression                                                                                                           | operator precedence                                       |
| 186 | choose correct declaration of overloaded inequality(!=) operator for class string as non member friend function                                                                | friend bool operator!=(const string &s1,const string &s2) |
| 187 | which of the following is most suitable advantage of information hiding                                                                                                        | simplifies the model                                      |
| 188 | a fan has wings, which type of relation exists between fan and Wings in this                                                                                                   | composition                                               |

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|     |                                                                                                         |                                                                                         |
|-----|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
|     | sentence                                                                                                |                                                                                         |
| 189 | multiple inheritance is implemented between class through----- symbol                                   | comma(,)                                                                                |
| 190 | postfix unary operator is implemented in C ++ using non Member function with arguments                  | two                                                                                     |
| 191 | Supposed derived class is inherited from base class what happens when a derived class object is created | the constructor of base class exact is executed before the constructor of derived class |
| 192 | assignment operator are                                                                                 | right associative                                                                       |
| 193 | a postfix unary operator is implemented in C ++ using number function with                              | 1 dummy int argument                                                                    |
| 194 | which of the following is true about exception functions                                                | all of the given option                                                                 |



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**MIDTERM EXAMINATION**  
**Spring 2010**

**Question No: 1 ( Marks: 1 ) - Please choose one**  
**Which part of an object exhibits its state?**

- ▶ **Data (Page 23)**
- ▶ Operations
- ▶ Any public part
- ▶ Any private part

**Question No: 2 ( Marks: 1 ) - Please choose one**  
**Inheritance is a way to**

- ▶ organize data.
- ▶ pass arguments to objects of classes.
- ▶ **add features to existing classes without rewriting them. (Page 27)**
- ▶ improve data-hiding and encapsulation.

**Question No: 3 ( Marks: 1 ) - Please choose one**  
**Suppose you have been given the following design,**

**"A person has a name, age, address and sex. You are designing a class to represent a type of person called a patient. This kind of person may be given a diagnosis, have a spouse and may be alive".**  
**Given that the person class has already been created, what of the following would be appropriate to include when you design the patient class?**

- ▶ **registration date and diagnosis**      [Click here for Detail](#)
- ▶ age and sex
- ▶ sex and diagnosis
- ▶ diagnosis and age

**Question No: 4 ( Marks: 1 ) - Please choose one**

**What problem(s) may occur when we copy objects without using deep copy constructor?**

- ▶ Dangling pointer
- ▶ Memory Leakage
- ▶ **All of the given (Page 147)**
- ▶ System crash

**Question No: 5 ( Marks: 1 ) - Please choose one**  
**this pointers are not accessible for static member functions.**

- ▶ **True (Page 114)**
- ▶ False

**Question No: 6 ( Marks: 1 ) - Please choose one**  
**A static member function cannot be declared.**

- ▶ Static
- ▶ Implicit
- ▶ Explicit
- ▶ **Virtual** [Click here for detail](#)

**Question No: 7 ( Marks: 1 ) - Please choose one**  
**\_\_\_\_\_ remain in memory even when all objects of a class have been destroyed.**

- ▶ **Static variables** (Page 111)
- ▶ Instance variable
- ▶ Primitive variables
- ▶ None of given

**Question No: 8 ( Marks: 1 ) - Please choose one**  
**Friend functions are \_\_\_\_\_ functions of a class.**

- ▶ None of given
- ▶ object member
- ▶ **non-member** (Page 136)
- ▶ data member

**Question No: 9 ( Marks: 1 ) - Please choose one**  
**\_\_\_\_\_, which means if A declares B as its friend it does NOT mean that A can access private data of B. It only means that B can access all data of A.**

- ▶ **Friendship is one way only**
- ▶ Friendship is two way only
- ▶ NO Friendship between classes
- ▶ Any kind of friendship

**Question No: 10 ( Marks: 1 ) - Please choose one**  
**The statement objA=objB; will cause a compiler error if the objects are of different classes.**

- ▶ **True**
- ▶ False

**Question No: 11 ( Marks: 1 ) - Please choose one**  
**Identify which of the following overloaded operator function's declaration is appropriate for the given call?**



**Rational\_number\_1 + 2.325**

**Where Rational\_number\_1 is an object of user defined class Rational\_number.**

- ▶ Rational\_number operator+( Rational\_number & obj);
- ▶ **Rational\_number operator+(double& obj); (Page 145)**
- ▶ Rational\_number operator+(Rational\_number &obj, double& num);
- ▶ operator+(double& obj);

**Question No: 12 ( Marks: 1 ) - Please choose one**

**Which operator can not be overloaded?**

- ▶ The relation operator ( >= )
- ▶ Assignment operator ( = )
- ▶ Script operator ( [] )
- ▶ **Conditional operator ( ? : ) (Page 141)**

**Question No: 13 ( Marks: 1 ) - Please choose one**

**To convert from a user-defined class to a basic type, you would most likely use**

- ▶ a built-in conversion operator.
- ▶ a one-argument constructor.
- ▶ an overloaded = operator.
- ▶ **a conversion operator that's a member of the class.**

**Question No: 14 ( Marks: 1 ) - Please choose one**

**The technique in which we visualize our programming problems according to real life's problems is called**

- 
- ▶ structured programming
  - ▶ **object oriented Programming (Page 9)**
  - ▶ procedural programming
  - ▶ non of the given

**Question No: 15 ( Marks: 1 ) - Please choose one**

**In object orientated programming, a class of objects cans \_\_\_\_\_ properties from another class of objects**

- ▶ Utilize
- ▶ Borrow
- ▶ **Inherit [Click here for detail](#)**
- ▶ Adopt

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**Question No: 16 ( Marks: 1 ) - Please choose one**

**A C++ class is similar to -----**

- ▶ **Structure**                      [Click here for detail](#)
- ▶ Header File
- ▶ Library File
- ▶ None of the given

**MIDTERM EXAMINATION**  
**Spring 2010**

**Question No: 1 ( Marks: 1 ) - Please choose one**

**What problem(s) may occur when we copy objects without using deep copy constructor?**

- ▶ Dangling pointer
- ▶ Memory Leakage
- ▶ **All of the given**                      (Page 147)
- ▶ System crash

**Question No: 2 ( Marks: 1 ) - Please choose one**

**Suppose that the Test class does not have an overloaded assignment operator. What happens when an assignment a=b; is given for two Test objects a and b?**

- ▶ The automatic assignment operator is used
- ▶ **The copy constructor is used**                      [Click here for detail](#)
- ▶ Compiler error
- ▶ Run-time error

**Question No: 3 ( Marks: 1 ) - Please choose one**

**a'A static member function can be called, even when a class is not \_\_\_\_\_.**

- ▶ Declared
- ▶ Define
- ▶ **Instantiated**                      C++ How to Program, Fifth Edition(Page 871)
- ▶ Called

**Question No: 4 ( Marks: 1 ) - Please choose one**

**Identify which of the following overloaded operator function's declaration is appropriate for the given call?**

**Rational\_number\_1 + 2.325**

**Where Rational\_number\_1 is an object of user defined class Rational\_number.**

- ▶ Rational\_number operator+( Rational\_number & obj);
- ▶ **Rational\_number operator+(double& obj); (Page 145)**
- ▶ Rational\_number operator+(Rational\_number &obj, double& num);
- ▶ operator+(double& obj);

**Question No: 5 ( Marks: 1 ) - Please choose one**  
 \_\_\_\_\_ provide the facility to access the data member.

- ▶ **accesser function (Page 68)**
- ▶ private function
- ▶ inline function
- ▶ None of the given

**Question No: 6 ( Marks: 1 ) - Please choose one**  
 Constant objects cannot change their state,

- ▶ **True (Page 105)**
- ▶ False

**Question No: 7 ( Marks: 1 ) - Please choose one**  
 The \_\_\_\_\_ relationship indicates that an object contains other objects.

- ▶ **None of given (Page 53)**
- ▶ 'has-a'
- ▶ 'is-a'
- ▶ 'be-

**Question No: 8 ( Marks: 1 ) - Please choose one**  
 Which one of the following features of OOP is used to derive a class from another?

- ▶ Encapsulation
- ▶ Polymorphism
- ▶ Data hiding
- ▶ **Inheritance (Page 25)**

**Question No: 9 ( Marks: 1 ) - Please choose one**  
 \_\_\_\_\_ is a relationship

- ▶ **Inheritance (Page 25)**
- ▶ Polymorphism
- ▶ abstraction
- ▶ encapsulation



**Question No: 10 ( Marks: 1 ) - Please choose one**  
**\_\_\_\_\_ satisfy the condition of polymorphism**

- ▶ Carbon
- ▶ Diamond
- ▶ Coal
- ▶ **all of the given** (Page 56)

**Question No: 11 ( Marks: 1 ) - Please choose one**  
**A generalization-specialization relation between classes are implemented using**

- ▶ data hiding
- ▶ friend classes
- ▶ encapsulation
- ▶ **inheritance** (Page 49)

**Question No: 12 ( Marks: 1 ) - Please choose one**  
**The >= operator can be overloaded.**

- ▶ **True** (Page 140)
- ▶ False

**Question No: 13 ( Marks: 1 ) - Please choose one**  
**In order to free the memory occupied by the object, we use -----**

- ▶ Constructor
- ▶ **Destructor** (Page 92)
- ▶ Shallow Copy
- ▶ Deep Copy

**Question No: 14 ( Marks: 1 ) - Please choose one**  
**Which of the following is not an example of multiple inheritances?-----**

- ▶ Mermaid
- ▶ **Woman** (Lecture No.5)
- ▶ None of the given
- ▶ Amphibious Vehicle

**Question No: 15 ( Marks: 1 ) - Please choose one**  
**Static variable can be initialized more than once.**

- ▶ **True** [Click here for detail](#)
- ▶ False

**Question No: 16 ( Marks: 1 ) - Please choose one**

**A generic class showing all the common attributes and a behavior of other classes represents a very important feature in oop called -----**

- ▶ **Inheritance** (Page 26)
- ▶ Encapsulation
- ▶ Polymorphism
- ▶ Abstraction

**MIDTERM EXAMINATION**  
**Spring 2010**

**Question No: 1 ( Marks: 1 ) - Please choose one**

**We can get only one unique value which can be used by all the objects of that class by the use of,**

- ▶ **static variables** [Click here for detail](#)
- ▶ dynamic variables
- ▶ instance variables
- ▶ data members

**Question No: 2 ( Marks: 1 ) - Please choose one**

**A member function having the same name as that of a class and a ~ sign with it is called,**

- ▶ Constructor
- ▶ Getter
- ▶ Setter
- ▶ **Destructor** (Page 92)

**Question No: 3 ( Marks: 1 ) - Please choose one**

**Using encapsulation we can achieve**

- ▶ **Information hiding** (Page 16)
- ▶ Least interdependencies among modules
- ▶ Implementation independence
- ▶ All of given options

**Question No: 4 ( Marks: 1 ) - Please choose one**

**Inheritance is a way to**

- ▶ **make general classes into more specific classes.** (Page 27)
- ▶ pass arguments to objects of classes.
- ▶ improve data hiding and encapsulation.
- ▶ providing class growth through natural selection.

**Question No: 5 ( Marks: 1 ) - Please choose one**  
**Static variable can be initialized more than once.**

True

**False (Page 107)**

**Question No: 6 ( Marks: 1 ) - Please choose one**

**For classes with common behavior, you can save effort by placing the common behavior in a \_\_\_\_\_.**

► Derived Class

► **Base class (Page 29)**

► Deprived Class

► Named class

**Question No: 7 ( Marks: 1 ) - Please choose one**

**Which of the following are an advantage of OOP?**

► OOP makes it easy to re-use the code

► It provides an ability to create one user defined data type by extending the other

► It provides the facility of defining Abstract data types through which real world entities can be defined better.

► **All of the given options**

[Click here for detail](#)

**Question No: 8 ( Marks: 1 ) - Please choose one**

**The >= operator can be overloaded.**

► **True (Page 140)**

► False

**Question No: 9 ( Marks: 1 ) - Please choose one**

**A static member function cannot be declared.**

► Static

► Implicit

► Explicit

► **Virtual** [Click here for detail](#)

**Question No: 10 ( Marks: 1 ) - Please choose one**

**Static variables act like a global variable in the context or scope of the class.**

► **True** [click here for detail](#)

► False

**Question No: 11 ( Marks: 1 ) - Please choose one**

**The compiler won't object if you overload the \* operator to perform division.**

► **True** [Click here for detail](#)

► False

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**Question No: 12 ( Marks: 1 ) - Please choose one**

**We can use "this" pointer in the constructor in the body and even in the initialization list of any class if we are careful,**

- ▶ TRUE
- ▶ **False**

**Question No: 13 ( Marks: 1 ) - Please choose one**

**A C++ class is similar to -----**

- ▶ **Structure** [Click here for detail](#)
- ▶ Header File
- ▶ Library File
- ▶ None of the given

**Question No: 14 ( Marks: 1 ) - Please choose one**

**Which operator can not be overloaded?**

- ▶ The relation operator ( >= )
- ▶ Assignment operator ( = )
- ▶ Script operator ( [] )
- ▶ **Conditional operator ( ? : ) (Page 141)**

**Question No: 15 ( Marks: 1 ) - Please choose one**

**An overloaded operator always requires one less argument than its number of operands.**

- ▶ **True (Page 896)**
- ▶ False

**Question No: 16 ( Marks: 1 ) - Please choose one**

**A generalization-specialization relation between classes are implemented using**

- ▶ data hiding
- ▶ friend classes
- ▶ encapsulation
- ▶ **inheritance (Page 49)**

**Question No: 17 ( Marks: 1 ) - Please choose one**

**In OOP a class is an example of \_\_\_\_\_**

- ▶ Data Type
- ▶ Abstract Type
- ▶ **User Defined Type (Page 66)**
- ▶ None of the given

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**Question No: 18 ( Marks: 1 ) - Please choose one**

**A class can be identified from a statement by -----**

▶ **Noun (Page 58)**

▶ Pronoun

▶ Verb

▶ Adverb

**Question No: 19 ( Marks: 1 ) - Please choose one**

**The members of a class that can be accessed without creating the object of the class is called**

▶ Private member

▶ Data Member

▶ **Public Member (How to Program page 983)**

▶ Static

**MIDTERM EXAMINATION**  
**Spring 2010**

**Question No: 1 ( Marks: 1 ) - Please choose one**

**Suppose there is an object of type Person, which of the following can be considered as one of its attributes**

▶ Name

▶ Age

▶ Work()

▶ **Both Name and Age (page 13)**

**Question No: 2 ( Marks: 1 ) - Please choose one**

**What a derived class can add?**

▶ New data members

▶ New member functions and New friend functions

▶ New constructors and destructor

▶ **All of given [Click here for Detail](#)**

**Question No: 3 ( Marks: 1 ) - Please choose one**

**\_\_\_\_\_ is/are used to access information hidden within an object?**

▶ Interface

▶ Private data members

▶ **Private member functions (Page 69)**

▶ Both public and private members



**Question No: 4 ( Marks: 1 ) - Please choose one**  
**this pointers are not accessible for static member functions.**

- ▶ **True** (Page 14)
- ▶ False

**Question No: 5 ( Marks: 1 ) - Please choose one**  
**A static member function cannot be declared.**

- ▶ Static
- ▶ Implicit
- ▶ Explicit
- ▶ **Virtual** [Click here for detail](#)

**Question No: 6 ( Marks: 1 ) - Please choose one**  
**C++ compiler does not allow to dynamically allocate memory for objects**

- ▶ **False** [Click here for detail](#)
- ▶ True

**Question No: 7 ( Marks: 1 ) - Please choose one**

**Given the following class**

```
class Base{
int Age=33;
}
```

**How you can improve above class with respect to accessing the field Age?**

- ▶ Define the variable Age as private
  - ▶ Define the variable Age as protected
  - ▶ **Define the variable Age as private and create a get method that returns it and a set method that updates it**
- [Click here for detail](#)
- ▶ Define the variable Age as protected and create a set method that returns it and a get method that updates it

**Question No: 8 ( Marks: 1 ) - Please choose one**  
**Friend class and friend function can be used as an alternate to each other**

- ▶ **True** [Click here for detail](#)
- ▶ False

**Question No: 9 ( Marks: 1 ) - Please choose one**  
**Which of the following operators always takes no argument if overloaded?**

- ▶ /
- ▶ -

- ▶ +
- ▶ ++

**Question No: 10 ( Marks: 1 ) - Please choose one**

**Suppose that the Test class does not have an overloaded assignment operator. What happens when an assignment  $a=b$ ; is given for two Test objects a and b?**

- ▶ The automatic assignment operator is used
- ▶ **The copy constructor is used** [Click here for detail](#)
- ▶ Compiler error
- ▶ Run-time error

**Question No: 11 ( Marks: 1 ) - Please choose one**

**Assume a class C with objects obj1, obj2, and obj3. For the statement  $obj3 = obj1 - obj2$  to work correctly, if the overloaded - operator must**

- ▶ take two arguments.
- ▶ **return a value** [Click here for detail](#)
- ▶ create a named temporary object.
- ▶ take four arguments

**Question No: 12 ( Marks: 1 ) - Please choose one**

**Which operator can not be overloaded?**

- ▶ The relation operator (  $\geq$  )
- ▶ Assignment operator ( = )
- ▶ Script operator ( [ ] )
- ▶ **Conditional operator ( ? : )** (Page 141)

**Question No: 13 ( Marks: 1 ) - Please choose one**

**We achieve independence of internal implementation from its external interface through-----**

- ▶ Encapsulation
- ▶ Information Hiding
- ▶ **Abstraction** [Click here for detail](#)
- ▶ both encapsulation and information hiding

**Question No: 14 ( Marks: 1 ) - Please choose one**

**Which one of the following is not an object association?**

- ▶ Simple Association
- ▶ **Inheritance** (Page 49)
- ▶ Aggregation
- ▶ Composition

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**Keeping in view the principle of abstraction, which of the above information the company needs to save as employee's record?**

- ▶ 2, 4
- ▶ 1, 3, 5
- ▶ **1, 2, 3 (Lecture No.3)**
- ▶ 1, 2, 3, 4

**Question No: 4 ( Marks: 1 ) - Please choose one**  
**What a derived class can add?**

- ▶ New data members
- ▶ New member functions and New friend functions
- ▶ New constructors and destructor
- ▶ **All of given**      [Click here for Detail](#)

**Question No: 5 ( Marks: 1 ) - Please choose one**  
**The concept of derived classes is involved in,**

- ▶ **inheritance**      (Page 25)
- ▶ encapsulation
- ▶ structure
- ▶ array

**Question No: 6 ( Marks: 1 ) - Please choose one**

**Your chief Software designer has shown you a sketch of the new Computer parts system she is about to create. At the top of the hierarchy is a Class called Computer and under this are two child classes. One is called LinuxPC and one is called WindowsPC. The main difference between the two is that one runs the Linux operating System and the other runs the Windows System (of course another difference is that one needs constant re-booting and the other runs reliably). Under the WindowsPC are two Sub classes one called Server and one Called Workstation. How might you appraise your designers work?**

- ▶ Give the goahead for further design using the current scheme
- ▶ **Ask for a re-design of the hierarchy with changing the Operating System to a field rather than**

**Class type**      [Click here for detail](#)

- ▶ Ask for the option of WindowsPC to be removed as it will soon be obsolete
- ▶ Change the hierarchy to remove the need for the superfluous Computer Class.

**Question No: 7 ( Marks: 1 ) - Please choose one**  
**Consider the code below,**

```
class Fred {
public:
Fred();
```

```

...
};
int main()
{
Fred a[10];
Fred* p = new Fred[10];
...
}

```

Select the best option,

- ▶ Fred a[10]; calls the default constructor 09 times
- Fred\* p = new Fred[10]; calls the default constructor 10 times
  - ▶ Produce an error
  - ▶ **Fred a[10]; calls the default constructor 11 times**
- Fred\* p = new Fred[10]; calls the default constructor 11 times**
  - ▶ Fred a[10]; calls the default constructor 10 times
- Fred\* p = new Fred[10]; calls the default constructor 10 times

**Question No: 8 ( Marks: 1 ) - Please choose one**  
**Which construct is the source for the creation of an object?**

- ▶ Destructor of the class
- ▶ **New operator** [Click here for detail](#)
- ▶ Delete operator
- ▶ Constructor of the class

**Question No: 9 ( Marks: 1 ) - Please choose one**  
**this pointers are not accessible for static member functions.**

- ▶ **True (Page 14)**
- ▶ False

**Question No: 10 ( Marks: 1 ) - Please choose one**  
**When a variable is define as static in a class then all object of this class,**

- ▶ Have different copies of this variable
- ▶ **Have same copy of this variable** [Click here for detail](#)
- ▶ Can not access this variable
- ▶ None of given

**Question No: 11 ( Marks: 1 ) - Please choose one**  
**\_\_\_\_\_ remain in memory even when all objects of a class have been destroyed.**

- ▶ **Static variables (Page 111)**
- ▶ Instance variable
- ▶ Primitive variables
- ▶ None of given



**Question No: 12 ( Marks: 1 ) - Please choose one**

**Given the following class**

```
class Base{
 int Age=33;
}
```

**How you can improve above class with respect to accessing the field Age?**

- ▶ Define the variable Age as private
- ▶ Define the variable Age as protected
- ▶ **Define the variable Age as private and create a get method that returns it and a set method that updates it** [Click here for detail](#)
- ▶ Define the variable Age as protected and create a set method that returns it and a get method that updates it

**Question No: 13 ( Marks: 1 ) - Please choose one**

**The life of sub object is not dependant on the life of master class in \_\_\_\_\_.**

- ▶ Composition
- ▶ **Aggregation** (Page 134)
- ▶ Separation
- ▶ None of the given

**Question No: 14 ( Marks: 1 ) - Please choose one**

**Which one is not keyword in C++?**

- ▶ operator
- ▶ **B\_op** [Click here for detail](#)
- ▶ const
- ▶ None of given

**Question No: 15 ( Marks: 1 ) - Please choose one**

**The >= operator can be overloaded.**

- ▶ **True** (Page 140)
- ▶ False

**Question No: 16 ( Marks: 1 ) - Please choose one**

**Identify which of the following overloaded operator function's declaration is appropriate for the given call?**

**Rational\_number\_1 + 2.325**

**Where Rational\_number\_1 is an object of user defined class Rational\_number.**

- ▶ Rational\_number operator+( Rational\_number & obj);
- ▶ **Rational\_number operator+(double& obj);**
- ▶ Rational\_number operator+(Rational\_number &obj, double& num);
- ▶ operator+(double& obj);

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**MIDTERM EXAMINATION**  
**Spring 2009**

**Question No: 1 ( Marks: 1 ) - Please choose one**

**A template provides a convenient way to make a family of**

- ▶ variables and data members
- ▶ **functions and classes**      [Click here for detail](#)
- ▶ classes and exceptions
- ▶ programs and algorithms

**Question No: 2 ( Marks: 1 ) - Please choose one**

**Which one of the following terms must relate to polymorphism?**

- ▶ Static allocation
- ▶ Static typing
- ▶ **Dynamic binding**    (How to program , page 1103)    [Click here for more detail](#)
- ▶ Dynamic allocation

**Question No: 3 ( Marks: 1 ) - Please choose one**

**What is true about function templates?**

- ▶ The compiler generates only one copy of the function template
- ▶ **The compiler generates a copy of function respective to each type of data**    (Page 257)
- ▶ The compiler can only generate copy for the int type data
- ▶ None of the given.

**Question No: 4 ( Marks: 1 ) - Please choose one**

**Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?**

- ▶ **Templates**      [Click here for detail](#)
- ▶ Overloading
- ▶ Data hiding
- ▶ Encapsulation

**Question No: 5 ( Marks: 1 ) - Please choose one**  
**template <>**

**class Vector<char\*> { }**

**This is an example of partial specialization.**

- ▶ **True**    (Page 281)
- ▶ False



**Question No: 6 ( Marks: 1 ) - Please choose one**

Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be concrete, while classes like Sphere and Cube would normally be abstract.

▶ True

▶ False [Click here for Detail](#)

**Question No: 7 ( Marks: 1 ) - Please choose one**

A non-virtual member function is defined in a base class and overridden in a derived class; if that function is called through a base-class pointer to a derived class object, the derived-class version is used.

▶ True

▶ False [Click here for Detail](#)

**Question No: 8 ( Marks: 1 ) - Please choose one**

Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access

▶ public members of Derv. [Click here for Detail](#)

▶ protected members of Derv.

▶ private members of Derv.

▶ protected members of Base.

**Question No: 9 ( Marks: 1 ) - Please choose one**

In order to define a class template, the first line of definition must be:

▶ `template <typename T>` (Page 281)

▶ `typename <template T>`

▶ `Template Class <ClassName>`

▶ `Class <Template T>`

**Question No: 10 ( Marks: 1 ) - Please choose one**

If there is a pointer p to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a nonvirtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the \_\_\_\_\_ class to be executed.

▶ Base

▶ Derived

▶ Abstract

▶ virtual (not sure)

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**Question No: 11 ( Marks: 1 ) - Please choose one**

**When the base class and the derived class have a member function with the same name, you must be more specific which function you want to call (using \_\_\_\_\_).**

- ▶ scope resolution operator
- ▶ dot operator
- ▶ null operator
- ▶ **Operator overloading** (Page 204) [Click here for Detail](#)

**Question No: 12 ( Marks: 1 ) - Please choose one**

**Non Template Friend functions of a class are friends of \_\_\_\_\_instance/s of that class.**

- ▶ **All** [Click here for detail](#)
- ▶ One specific
- ▶ All instances of one date type
- ▶ None of the given options

**Question No: 13 ( Marks: 1 ) - Please choose one**

**The find() algorithm**

- ▶ finds matching sequences of elements in two containers.
- ▶ finds a container that matches a specified container.
- ▶ **takes iterators as its first two arguments.** [Click here for Detail](#)
- ▶ takes container elements as its first two arguments.

**Question No: 14 ( Marks: 1 ) - Please choose one**

**If you define a vector v with the default constructor, and define another vector w with a one-argument constructor to a size of 11, and insert 3 elements into each of these vectors with push\_back(), then the size() member function will return \_\_\_\_\_ for v and \_\_\_\_\_ for w.**

- ▶ 11 for v and 3 for w.
- ▶ 0 for v and 0 for w.
- ▶ 0 for v and 3 for w.
- ▶ **3 for v and 11 for w.**

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**Question No: 15 ( Marks: 1 ) - Please choose one**

**Which of the following may not be an integral part of an object?**

- ▶ State
- ▶ Behavior
- ▶ Protected data members
- ▶ **All of given (not sure)**

**Question No: 16 ( Marks: 1 ) - Please choose one**

**Which is not the Advantage of inheritance?**

- ▶ providing class growth through natural selection.
- ▶ facilitating class libraries.
- ▶ **avoiding the rewriting of code.** [Click here for Detail](#)
- ▶ providing a useful conceptual framework.

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## CS304 Quiz No.2 Spring 2020 by MCS of Virtualallians

1. The parameters given in temple definition other than those used for mentioning templates types are called \_\_\_\_\_
  - a. Type Parameters
  - b. Non Type parameters
  - c. Default Type Parameters
  - d. None of the given option
2. Which of the following is the correct way to define a template class X?
  - a. Class < typename T>class X{}
  - b. Template class C{}
  - c. Template < typename T> classX{}
  - d. Typename<class T > class X{}
3. In resolution order of function template, compiler searches for \_\_\_\_\_ in the end
  - a. Complete specialization
  - b. Partial specialization
  - c. Ordinary function
  - d. Generic template
4. \_\_\_\_\_ may inherit from a complete specialization
  - a. Partial specialization
  - b. Complete specialization
  - c. Ordinary class
  - d. All of given options
5. Which of the following may inherit form the ordinary class?
  - a. Class template
  - b. Partial specialization
  - c. Complete specialization
  - d. All of given option
6. Which of the following is a sequence container?
  - a. Map
  - b. Vector
  - c. Set
  - d. multiset
7. Each \_\_\_\_\_ of a template class by default becomes function template
  - a. Date member
  - b. Member function
  - c. Type parameter
  - d. Object
8. When we specialize a function template, it is called \_\_\_\_\_
  - a. Function overloading
  - b. Function overriding
  - c. Function template overloading
  - d. Function template overriding

9. Which of the following is NOT an associative container?
- a. Set
  - b. Multiset
  - c. Map
  - d. Group
10. A \_\_\_\_\_ is a pointer which is declared outside the \_\_\_\_\_
- a. Container, cursor
  - b. Cursor, container
  - c. Container, class
  - d. This, cursor
11. Which of the following are collectively referred to as the first class containers?
- a. Associative containers and containers adapters
  - b. Sequence containers and associative containers
  - c. Sequence containers and container adapters
  - d. None of the given options
12. \_\_\_\_\_ class is a single class that provides functionality to operate on different type of data
- a. Friend
  - b. Template
  - c. Ordinary
  - d. None of the give options
13. Which of the following can be passed as type argument to template?
- a. Primitive type
  - b. User defined types
  - c. Both primitive type and user defined types
  - d. None of the given options
14. Which of the following is known as Dereference operator in C++?
- a. +
  - b. \*
  - c. &
  - d. ::
15. We can change behavior of template using \_\_\_\_\_
- a. Template parameters
  - b. Function parameters
  - c. Class Templates
  - d. none of the given options
16. Which of the following represents partial specialization?
- a. Template < int, char >
  - b. Template < class T, Type t>
  - c. Template<class T, Class U, int>
  - d. Template<typename T, class W>

17. When we declare template function as friend of any class, then which of the following specialization(s) of template function also become friend of the class granting friendship?

- a. Explicit
- b. Implicit
- c. Partial
- d. All of the given options

18. In statement “template <class T, class U, int I = 5>” then non-type parameters is

- a. Class T
- b. Class U
- c. int I
- d. All of the given option

19. Which of the following is the correct syntax for passing two type argument to a template?

- a. template < type T, type U>
- b. template < typename T, typename U>
- c. template <type T,U>
- d. template Typename < T,U >

20. Which of the following represents complete specialization?

- a. Template < int, char >
- b. Template < class T, float>
- c. Template<class T, Class U , int>
- d. Template< class , class W>

21. Which of the following is/are the key component of standard template library?

- a. Containers
- b. Iterators
- c. Algorithm
- d. All of the given option

22. Which of the following statement is true about partial specialization?

- a. Function templates cannot have partial specialization
- b. Class template cannot have partial specialization
- c. Both class templates and function template can have partial specialization.
- d. None of the given

23. Which of the following is correct code portion to initialize static data member “value” of int type for class X?

- a. Template <class T> X::value =0;
- b. Template<class T>int X::value =0;
- c. Template<class T> int X < T >::value = 0;
- d. Template<class T> X < T >::value =0;

24. When we want to have exactly identical operations on different data type, \_\_\_\_\_ are used

- a. Function Template



- b. Function Overloading
  - c. Function Overriding
  - d. None of the given option
25. In order to define a class template the first line of definition must be:
- a. **Template <typename T>**
  - b. typename < template T>
  - c. Template class <ClassName>
  - d. Class <Template T>
26. In case of template specialization, if compiler cannot find required complete specialization then it searches for some \_\_\_\_\_
- a. General template
  - b. Complete template
  - c. **Partial specialization**
  - d. None of the given
27. Which of the following is the correct code to instantiate the object of given template vector class of int type?
- ```
Template <class T>  
class vector  
{  
};
```
- a. Vector obj <int>;
 - b. Vector obj <> int;
 - c. **Vector <int>obj;**
 - d. Vector int obj;
28. Which of the following types of container is NOT provided by standard Template Library?
- a. Sequence containers
 - b. **Structured containers**
 - c. Associative containers
 - d. Container adapters
29. A class template _____
- a. **Facilitates reuse of class**
 - b. Does not facilitates reuse of class
 - c. Does not support generic methods
 - d. Does not support static members
30. Suppose a template class “test” has a static data members, how many copies of this static data member will be created when template class is instantiated by writing following line of C++ code?
- ```
Test <char>x,y,z
```
- a. Zero
  - b. One
  - c. Two
  - d. **Three**

31. STL stands for:

- a. Stack Template Library
- b. Standard Template Library
- c. Structured Template Library
- d. Secure Template Library

32. Which of the following is not a sequence container?

- a. Vector
- b. Deque
- c. Map
- d. List

33. Suppose a template class “test” has a static data members, how many copies of this static data member will be created when template class is instantiated by writing following line of C++ code?

```
Test <int>x;
```

```
Test <char>y;
```

- a. Zero
- b. One
- c. Two
- d. Three

34. Suppose we have a template class “Test” which have a static data member “data” which is initialized with 0. What will be output when following code is written in main() function for given class?

```
Test<int> a, b;
```

```
a.data = 5;
```

```
b.data = 7;
```

```
cout<<a.data;
```

- a. 0
- b. 1
- c. 5
- d. 7

35. In resolution order, highest priority is given to \_\_\_\_\_ in template specialization.

- a. General template
- b. Complete specialization
- c. Partial specialization
- d. None of the given options

36. Which of the following is a constrained version of some first-class container?

- a. Sequence containers
- b. Associative containers
- c. Container Adapter
- d. None of the given options

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Best of luck!





## Question # 2 of 30 ( Start time: 05:20:04 PM, 28 December 2020 )

Total Marks: 1

An instance of user defined type is called \_\_\_\_\_

Select the correct option

|                                  |          |
|----------------------------------|----------|
| <input checked="" type="radio"/> | Object   |
| <input type="radio"/>            | Class    |
| <input type="radio"/>            | Variable |
| <input type="radio"/>            | Program  |

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## Question # 17 of 30 ( Start time: 05:33:05 PM, 28 December 2020 )

Total Marks: 1

The process of extracting common characteristics from two or more classes and combining them into a generalized superclass is called \_\_\_\_\_.

Select the correct option

|                                  |                |    |
|----------------------------------|----------------|----|
| <input checked="" type="radio"/> | Generalization | // |
| <input type="radio"/>            | Specialization | // |
| <input type="radio"/>            | Abstraction    | // |
| <input type="radio"/>            | Composition    |    |

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Question # 18 of 30 ( Start time: 05:33:57 PM, 28 December 2020 )

Total Marks: 1

The sentence "Object Oriented Programming book in bookshelf" is an example of:

Select the correct option

- |                                  |                      |    |
|----------------------------------|----------------------|----|
| <input type="radio"/>            | Binary association   | // |
| <input type="radio"/>            | Association          | // |
| <input type="radio"/>            | Multiple association | // |
| <input checked="" type="radio"/> | Aggregation          |    |

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Question # 14 of 30 ( Start time: 05:30:06 PM, 28 December 2020 )

Total Marks: 1

Specialization means that derived class is behaviorally \_\_\_\_\_ with the base class.

Select the correct option

|                                  |              |
|----------------------------------|--------------|
| <input type="radio"/>            | Same         |
| <input type="radio"/>            | Compatible   |
| <input type="radio"/>            | Different    |
| <input checked="" type="radio"/> | Incompatible |

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Question # 6 of 30 ( Start time: 05:24:21 PM, 28 December 2020 )

Total Marks: 1

Which of the following operator doesn't take any argument if overloaded?

Select the correct option

|                                  |                           |
|----------------------------------|---------------------------|
| <input type="radio"/>            | /                         |
| <input type="radio"/>            | -                         |
| <input type="radio"/>            | +                         |
| <input checked="" type="radio"/> | None of the given choices |

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Question # 29 of 30 ( Start time: 05:42:08 PM, 28 December 2020 )

Total Marks: 1

The \_\_\_\_\_ tells the compiler what task the function will be performing.

Select the correct option

|                                  |                      |    |
|----------------------------------|----------------------|----|
| <input checked="" type="radio"/> | Function declaration | // |
| <input type="radio"/>            | Function calling     | // |
| <input type="radio"/>            | Function definition  |    |
| <input type="radio"/>            | None of the given    |    |

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Question # 20 of 30 ( Start time: 05:35:36 PM, 28 December 2020 )

Total Marks: 1

Statement "Yasir is a friend of Ali and Ali is a friend of Yasir" is an example of :

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Select the correct option

|                                  |                     |   |
|----------------------------------|---------------------|---|
| <input type="radio"/>            | One way association | / |
| <input checked="" type="radio"/> | Two way association | / |
| <input type="radio"/>            | Ternary association | / |
| <input type="radio"/>            | N-ary association   | / |

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Question # 25 of 30 ( Start time: 05:39:03 PM, 28 December 2020 )

Total Marks: 1

\_\_\_\_\_ is automatically called when the object is created.

Select the correct option

|                                  |                   |    |
|----------------------------------|-------------------|----|
| <input type="radio"/>            | member function   | // |
| <input type="radio"/>            | object            | // |
| <input checked="" type="radio"/> | constructor       |    |
| <input type="radio"/>            | None of the given |    |

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Question # 30 of 30 ( Start time: 05:43:17 PM, 28 December 2020 )

Total Marks: 1

In Object Oriented programming, objects communicate with each other through \_\_\_\_\_

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Select the correct option

|                                  |              |   |
|----------------------------------|--------------|---|
| <input type="radio"/>            | Messages     | / |
| <input type="radio"/>            | Data members | / |
| <input type="radio"/>            | Voice chat   | / |
| <input checked="" type="radio"/> | Ports        | / |

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Question # 26 of 30 ( Start time: 05:39:27 PM, 28 December 2020 )

Total Marks: 1

\_\_\_\_\_ associates the objects of the exact three classes.

Select the correct option

|                                  |                     |   |
|----------------------------------|---------------------|---|
| <input type="radio"/>            | Binary Association  | / |
| <input type="radio"/>            | Two way Association | / |
| <input checked="" type="radio"/> | Ternary Association |   |
| <input type="radio"/>            | N-ary Association   |   |

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Question # 11 of 30 ( Start time: 05:27:42 PM, 28 December 2020 )

Total Marks: 1

\_\_\_\_\_ is creating objects of one class inside another class.

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Select the correct option

|                                  |             |    |
|----------------------------------|-------------|----|
| <input type="radio"/>            | Association | // |
| <input checked="" type="radio"/> | Composition | // |
| <input type="radio"/>            | Aggregation | // |
| <input type="radio"/>            | Inheritance |    |

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Question # 12 of 30 ( Start time: 05:28:00 PM, 28 December 2020 )

Total Marks: 1

\_\_\_\_\_ is the reverse process of Generalization.

Select the correct option

|                                  |                |    |
|----------------------------------|----------------|----|
| <input type="radio"/>            | Aggregation    | // |
| <input type="radio"/>            | Inheritance    | // |
| <input type="radio"/>            | Dependence     | // |
| <input checked="" type="radio"/> | Specialization |    |

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Question # 28 of 30 ( Start time: 05:41:23 PM, 28 December 2020 )

Total Marks: 1

Which of the following allows to reuse characteristics of more than one parent class?

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Select the correct option

|                                  |                      |   |
|----------------------------------|----------------------|---|
| <input type="radio"/>            | Single inheritance   | / |
| <input checked="" type="radio"/> | Multiple Inheritance | / |
| <input type="radio"/>            | Polymorphism         | / |
| <input type="radio"/>            | Specialization       | / |

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| SR NO | MCQS QUESTION                                                                                                                   | ANSWER                                                  |
|-------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 1     | In expression $c1*c2+c3-c4$ which of the following will be executed in first order                                              | $C1*c2$                                                 |
| 2     | Inheritance is a way to                                                                                                         | Add features to existing classes without rewriting them |
| 3     | ----- represent "IS A" relationship                                                                                             | Inheritance                                             |
| 4     | Constructor without any parameter is called:                                                                                    | Default constructor                                     |
| 5     | Which of the following stores the address of a variable?                                                                        | Pointer                                                 |
| 6     | A Post-fix unary operator is implemented in C++ using Member function with                                                      | 1 dummy int argument                                    |
| 7     | minimum classes required in a program for implementing multiple inheritance                                                     | 3                                                       |
| 8     | Which of the following statement(s) describe constant member functions?                                                         | All of the given option                                 |
| 9     | composition is a strong relation relationship because                                                                           | Composed object cant exist independency                 |
| 10    | How many interactions are there in the following scenario Imran has the car. he drive the car and car has four wheels and doors | 3                                                       |
| 11    | if we extend our model ,and the rest of the model is not affected then it is called-----                                        | Flexibility                                             |
| 12    | choose correct declaration of overload*=1 operator for class complex as Member function                                         | Complex and operator*=(const complex & c)               |
| 13    | Consider the following class A{int a,b,c; public: A():B(10),C(0)a(7) };                                                         | A,b,c                                                   |

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|    |                                                                                                                                     |                                            |
|----|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| 14 | Class is a blue print of                                                                                                            | Objects                                    |
| 15 | Suppose we have a class student which of the following statement is correctly declaring the pointer to the object of student class? | Student*object                             |
| 16 | In c++ composition is relationship between -----and-----objects                                                                     | Whole, part                                |
| 17 | Which of the following Depicts the proper definition of class?                                                                      | a class is description of a kind of object |
| 18 | in class attributes are represented by ---- and----- respectively                                                                   | data member, Member function               |
| 19 | which of the following represents the two Association                                                                               | Employee works for the company             |
| 20 | Entities in OO model are represented by                                                                                             | Rectangle                                  |
| 21 | Which of the following features of OOP is used to derive a class from another?                                                      | Inheritance                                |
| 22 | In C ++ which of the following is defined as a stream insertion operator?                                                           | <<                                         |
| 23 | choose the correct declaration of overloaded equality (==)operator for class string as a Member function                            | Bool operator==(const string and s)        |
| 24 | ---- is the relationship between the part object and the whole object                                                               | Composition                                |
| 25 | object hides its information from others this concept is known as                                                                   | information hiding                         |
| 26 | the concept of derived classes is involved in                                                                                       | Inheritance                                |
| 27 | abstraction provides information according to                                                                                       | user perspective                           |
| 28 | which of the following is not an object Association                                                                                 | Inheritance                                |
| 29 | which of the following classes are used by amphibious vehicle to inherit characteristics                                            | both land and water vehicle                |

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|    |                                                                                                             |                                                              |
|----|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| 29 | object data includes                                                                                        | both states and attributes of an object                      |
| 30 | choose correct declaration of overloaded inequality operator for class string as non member friend function | friend pool operator<br>!=(const string &s1,const string &2) |
| 31 | How many objects of a given class may be constructed in an application?                                     | as many as the application ask for                           |
| 32 | how many objects of a given class may be constructed in an application                                      | as many as the application ask for                           |
| 33 | Which of the following statements describe constant member functions?                                       | all of the given option                                      |
| 34 | student *ptr=new student(" Ali"); is an example of                                                          | static memory allocation                                     |
| 35 | in----- base class can be replaced by the derived class                                                     | extension                                                    |
| 36 | Which of the following is a correct way of declaring constant variable ROLLNO?                              | cosnst int Roll no;                                          |
| 37 | which of the following operator doesn't take argument hit overloaded                                        | none of the given choices                                    |
| 38 | which of the following is part of the Class diagram                                                         | all of the given option                                      |
| 39 | a child inherits characteristic from its                                                                    | parent                                                       |
| 40 | which of the following issues can be caused by the use of friend function                                   | programming Bugs                                             |
| 41 | which of the following is not true about 'constant number function                                          | all of the given option                                      |
| 42 | Which one is not the main features of object oriented programming?                                          | exception handling                                           |
| 43 |                                                                                                             |                                                              |
| 44 | in object oriented programming, objects communicate with each other through                                 | messages                                                     |
| 45 | static data member is declared                                                                              | inside the class                                             |
| 46 | the concept of derived classes is involved in                                                               | inheritance                                                  |

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|    |                                                                                                                                               |                                                                |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| 47 | Which of the following can we attribute of an object "Ali"?                                                                                   | address                                                        |
| 48 | Which of the following keyword tell the compiler to substitute the code within the function definition for every instance of a function call? | inline                                                         |
| 49 | inheritance is the relationship between                                                                                                       | classes                                                        |
| 50 | how many objects are involved in the binary Association                                                                                       | 2                                                              |
| 51 | all objects of same class has ----- attributes                                                                                                | same                                                           |
| 52 | which of the following concept is more close to encapsulation                                                                                 | information hiding                                             |
| 53 | member initialization list is used to                                                                                                         | initialize data member                                         |
| 54 | aggregation between objects show that                                                                                                         | objects have weak relationship                                 |
| 55 | there is /are ways to create user defined types for objects in C ++                                                                           | two                                                            |
| 56 | composition is a strong relationship because:                                                                                                 | composed object can not exist independently                    |
| 57 | in polymorphism messenger can be interpreted in different ways depending upon the----- class                                                  | receiver                                                       |
| 58 | which of the following is most suitable advantage of information hiding                                                                       | simplifies the model                                           |
| 59 | the process of extracting common characteristics from two or More classes and combining them into a generalized superclass is called          | Generalization                                                 |
| 60 | ----- operator is used to create objects at run time                                                                                          | new                                                            |
| 61 | choose the correct option to declare an overloaded Stream insertion operator for                                                              | friend ostream and operator<<(ostream and os,const string & s) |

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|    |                                                                                                                    |                               |
|----|--------------------------------------------------------------------------------------------------------------------|-------------------------------|
|    | userdefined class "string" as a non-member friend function                                                         |                               |
| 62 | static data member is declared                                                                                     | inside the class              |
| 63 | an instance of user defined type is called                                                                         | object                        |
| 64 | which of the following is true about constructor                                                                   | all of the given option       |
| 65 | Which of the following correct way of declaring constant variable ROLLNO?                                          | const int ROLLNO;             |
| 66 | inheritance is shown by                                                                                            | filled arrow                  |
| 67 | which of the following operators operate on one operand                                                            | unary operator                |
| 68 | which of the following will happen when a constant function tried to change the value of data members of the class | compile time error will occur |
| 69 | in object oriented programming, objects communicate with each other through---                                     | messages                      |
| 70 | How the information hidden within an object can be accessed?                                                       | through its interface         |
| 71 | object orientation approach can be achieved using object and their-----                                            | interactions                  |
| 72 | constructors have ----- return type                                                                                | no                            |
| 73 | which of the following is an example of unary operator                                                             | increment operator(++)        |
| 74 | memory is allocated to non-static members only when                                                                | object is created             |
| 75 | 'in class B inherit from class A which of the following is child class                                             | class B                       |
| 76 | the----- tells the compiler what task the function will be performing                                              | function definition           |
| 77 | Consider the statement" room has chair" which of the following type of Association exists between whom and chair?  | aggregation                   |

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|    |                                                                                                           |                                                      |
|----|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| 78 | how many objects are involved in the N-ary Association                                                    | more than 3                                          |
| 79 | deconstruct is a function which has the same name as that of class but starts with a----- sign            | ~                                                    |
| 80 | which of the following is a tangible entity                                                               | car                                                  |
| 81 | member initializer list is used to                                                                        | initialize data members                              |
| 82 | we can get the address of variable stored in pointer using-----                                           | & symbol                                             |
| 83 | two sentence object oriented programming book in bookshelf is an example of                               | aggregation                                          |
| 84 | which of the following best defines the obstruction                                                       | hide the implementation and show current perspective |
| 85 | which of the following define default constructor 5 class student                                         | student(){//..}                                      |
| 86 | object data includes                                                                                      | both state and attributes of an object               |
| 87 | minimum classes required in a program for implementing multiple inheritance                               | 3                                                    |
| 88 | information hiding can be achieved through-----                                                           | encapsulation, abstraction                           |
| 89 | compiler generated defaulter structure is called----- hand user written default constructor is called---- | implicit, explicit                                   |
| 90 | which of the following is true about constructor                                                          | all of the given option                              |
| 91 | In which of the following different forms office single entity exist?                                     | polymorphism                                         |
| 92 | data members of a class student can be:                                                                   | all of the given option                              |
| 93 | the----- turn off the compiler what does the function will be performing                                  | function definition                                  |



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|     |                                                                                                                                              |                                                                                                  |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| 94  | Suppose a class does not have any constructor what will happen when an object of this class is created?                                      | compiler will call implicit default constructor                                                  |
| 95  | how can we differentiate between constructor and destructor                                                                                  | destructor are preceded with a Tilda(~) symbol, and constructor are not preceded with any symbol |
| 96  | ---- is used for hiding the details of an object from the other parts of a program                                                           | encapsulation                                                                                    |
| 97  | Class diagram include                                                                                                                        | class name, attributes, operation                                                                |
| 98  | through interface was access objects-----<br>--                                                                                              | behavior                                                                                         |
| 99  | composition is a strong relationship because                                                                                                 | compose object cannot exist independently                                                        |
| 100 | copy constructor is called when:                                                                                                             | an object is created in term of Preexisting object                                               |
| 101 | choose the correct option to declare an overloaded stream insertion operator for a user defined class string as a non-Member friend function | friend ostream & operator(ostream &os.const string &s)                                           |
| 102 | a post-fix unary operator implemented in C plus plus using non Member function with---- arguments                                            | two                                                                                              |
| 103 | in class attributes and behaviors are represented by---- and--- respectively                                                                 | data members, member functions                                                                   |
| 104 | which of the following best describes the relationship between book and book chapter classes                                                 | composition                                                                                      |
| 105 | which of the following is true about destructor                                                                                              | all of the given option                                                                          |
| 106 | in composition----- are called from composing objects to composed subjects                                                                   | destructors                                                                                      |

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|     |                                                                                             |                                            |
|-----|---------------------------------------------------------------------------------------------|--------------------------------------------|
| 107 | which of the following can be used an alternative to Static members                         | Global variables                           |
| 108 | which of the following statement is true about static function of a class                   | it is used to access static data members   |
| 109 | which of the following operator does not take any arguments if overloaded                   | none of the given choices                  |
| 110 | what is the output of the following code<br>int main(){int cost<br>x=10;cout<<++x;return0;} | error                                      |
| 111 | abstraction includes-----                                                                   | relevant information                       |
| 112 | Which of the following can be the behavior of an object "Usman"?                            | eat                                        |
| 113 | a class defined within another class is                                                     | nested class                               |
| 114 | when an object is initialized, is automatically called by the compiler                      | constructor                                |
| 115 | the compiler treats the Member function of a class as                                       | inline function                            |
| 116 | an object has attributes operation ,and---<br>----                                          | unique identity                            |
| 117 | Which of the following is not an access specifier in C ++?                                  | hidden                                     |
| 118 | which of the following parts of an object exhibits its state                                | data                                       |
| 119 | ---- is the relationship between the part object and the whole object                       | composition                                |
| 120 | how can we identify classes from a given problem statement                                  | by extracting nouns from problem statement |
| 121 | what "friend" should be placed in                                                           | function declaration                       |
| 122 | generalization is----- approach                                                             | bottom-up                                  |
| 123 | static data members can be accessed through-----                                            | dot operator and scope resolution operator |
| 124 | composition age----- and aggregation is---<br>---- relationship                             | strong, weak                               |

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|     |                                                                                                                                                                |                                                               |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| 125 | suppose there is an object of type person, which of the following can be considered as one of its attributes                                                   | name                                                          |
| 126 | Which of the following depicts the proper definition of class?                                                                                                 | a class is a description of a kind of object                  |
| 127 | a child inherits characteristics from its-----<br>--                                                                                                           | parent                                                        |
| 128 | the car is composed of wheels what is the relationship between car and wheels?                                                                                 | composition                                                   |
| 129 | ----- is creating objects of one class inside another class                                                                                                    | composition                                                   |
| 130 | the other name of specialization is-----                                                                                                                       | restriction                                                   |
| 131 | suppose a is an int type static data member class test then which of the following is correct way of initialization the static variable outside the class test | int test::a=0                                                 |
| 132 | Which of the following stores the address of variable?                                                                                                         | Pointer                                                       |
| 133 |                                                                                                                                                                |                                                               |
| 134 |                                                                                                                                                                |                                                               |
| 135 | static data member is declared                                                                                                                                 | inside the class                                              |
| 136 | polymorphism makes the system                                                                                                                                  | all of the given option                                       |
| 137 | which of the following statement best describes the constructor                                                                                                | constructor is used to initialize the data members of a class |
| 138 | data member(s) of a class student can be                                                                                                                       | all of the given option                                       |
| 139 | which of the following is a strong relationship                                                                                                                | composition                                                   |
| 140 | all objects of same class has----- attributes                                                                                                                  | same                                                          |
| 141 | in C ++ composition is a relationship between----and---- objects                                                                                               | whole, part                                                   |

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|     |                                                                                                                                                                                                                                                      |                                                           |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| 142 | suppose 12 class string assignment operator is overloaded with following declaration void operator=(const string &); what will happen when we will write following statement in main() str=str2=str3;where,str1,str2,str3 are object of class string | compiler will generate compile time error                 |
| 143 | in class B inherits from class A then it contains all characteristics of----                                                                                                                                                                         | class A                                                   |
| 144 | which of the following is not an example of multiple inheritance                                                                                                                                                                                     | car                                                       |
| 145 | which of the following can be created without any object of a class?                                                                                                                                                                                 | static data members                                       |
| 146 | consider the following code segment<br>class test{int a;int b;int c;}                                                                                                                                                                                | junk value,5, junk value                                  |
| 147 | object orientation approach can be achieved using objects and their-----                                                                                                                                                                             | interactions                                              |
| 148 | member functions defined inside a class declaration are----- by default                                                                                                                                                                              | private                                                   |
| 149 | how can we identify classes from a given problem statement                                                                                                                                                                                           | by extracting nouns from problem statement                |
| 150 | which of the following statement is not true about static variable of a class?                                                                                                                                                                       | static variable belongs to particular instance of a class |
| 151 | there are----- types of constructor in C++                                                                                                                                                                                                           | 3                                                         |
| 152 | statement yasir is a friend of Ali and Ali is a friend of yasir is an example of                                                                                                                                                                     | two way Association                                       |
| 153 | this pointer cannot be passed implicitly to----- functions                                                                                                                                                                                           | static member                                             |
| 154 | which of the following issues can be caused by the use of friend function                                                                                                                                                                            | all of the given                                          |

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|     |                                                                                                                         |                                           |
|-----|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| 155 | in----- send a message does not need to know exact class of the receiver                                                | none of the given option                  |
| 156 | minimum classes required in a program for implementing multiple inheritance                                             | 3                                         |
| 157 | -----Associates the object of the exact two classes                                                                     | binary Association                        |
| 158 | main characteristics of information hiding is                                                                           | keep the model simple                     |
| 159 | static data member is declared                                                                                          | inside the class                          |
| 160 | in expression $b=c/d$ , the arity of division(/) operator is                                                            | 2                                         |
| 161 | we can allocate dynamic memory in C ++ through operator                                                                 | new                                       |
| 162 | Which of the following is true about accessor functions?                                                                | all of the given option                   |
| 163 | in---- base class cannot always be replaced by the derived class                                                        | specialization                            |
| 164 | abstraction help to                                                                                                     | reduce the complexity                     |
| 165 | relationship in which child object gets destroyed if parent object is destroyed                                         | composition                               |
| 166 | object oriented programming is better than procedural programming because:                                              | it is more close to reality               |
| 167 | The derived class inherits behavior from the base class and also exhibits its own behavior. this concept is known as--- | Overriding                                |
| 168 | the other name of subtyping is                                                                                          | Extension                                 |
| 169 | A good model is----- related to life problem                                                                            | closely                                   |
| 170 | in generalization general class represents the ----- behavior of the class                                              | common                                    |
| 171 | how can we identify from a given problem statement?                                                                     | by extracting noun from problem statement |
| 172 | which of the following is not a access specifier in C ++                                                                | hidden                                    |

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|     |                                                                                                                                                                                 |                                                          |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| 173 | in composition----- are called from composing objects to composed objects                                                                                                       | destructors                                              |
| 174 | a class have only---- destructor                                                                                                                                                | one                                                      |
| 175 | suppose you have following C + + statement<br>int old value=10; int new value=++old value; what will be the value of old value and new Value after executing above statements ? | oldVlue=11,newVlue=11                                    |
| 176 | in constant Member function the type of this pointer is                                                                                                                         | constant pointer to constant data                        |
| 177 | ----- is creating objects of one class inside another class                                                                                                                     | composition                                              |
| 178 | ----- Associates the objects of the exact three classes                                                                                                                         | ternary Association                                      |
| 179 | generalization is implemented through                                                                                                                                           | inheritance                                              |
| 180 | which of the following is strong relationship                                                                                                                                   | composition                                              |
| 181 | composition is a strong relationship because                                                                                                                                    | composed object cannot exist independently               |
| 182 | ----- is set of function of an object exposed of other objects                                                                                                                  | interface                                                |
| 183 | suppose student is a class, which of the following defines constructor with one parameter for class student                                                                     | student(char*aName);                                     |
| 184 | which of the following is not true about constant Member function                                                                                                               | all of the given option                                  |
| 185 | ----- define the order of evaluation of an operator is an expression                                                                                                            | operator precedence                                      |
| 186 | choose correct declaration of overloaded inequality(!=) operator for class string as non member friend function                                                                 | friend bool operator!=(const string &s1,cont string &s2) |
| 187 | which of the following is most suitable advantage of information hiding                                                                                                         | simplifies the model                                     |

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|     |                                                                                                         |                                                                                         |
|-----|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| 188 | a fan has wings, which type of relation exists between fan and Wings in this sentence                   | composition                                                                             |
| 189 | multiple inheritance is implemented between class through----- symbol                                   | comma(,)                                                                                |
| 190 | postfix unary operator is implemented in C ++ using non Member function with arguments                  | two                                                                                     |
| 191 | Supposed derived class is inherited from base class what happens when a derived class object is created | the constructor of base class exact is executed before the constructor of derived class |
| 192 | assignment operator are                                                                                 | right associative                                                                       |
| 193 | a postfix unary operator is implemented in C ++ using number function with                              | 1 dummy int argument                                                                    |
| 194 | which of the following is true about exception functions                                                | all of the given option                                                                 |

## Question No: 1

Which part of an object exhibits its state?

- ▶ **Data (Page 23)**
- ▶ Operations
- ▶ Any public part
- ▶ Any private part

## Question No: 2

Inheritance is a way to



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- ▶ organize data.
- ▶ pass arguments to objects of classes.
- ▶ **add features to existing classes without rewriting them. (Page 27)**
- ▶ improve data-hiding and encapsulation.

## Question No: 3

Suppose you have been given the following design,

"A person has a name, age, address and sex. You are designing a class to represent a type of person called a patient. This kind of person may be given a diagnosis, have a spouse and may be alive".

Given that the person class has already been created, what of the following would be appropriate to include when you design the patient class?

- ▶ **registration date and diagnosis**
- ▶ age and sex
- ▶ sex and diagnosis
- ▶ diagnosis and age

## Question No: 4

What problem(s) may occur when we copy objects without using deep copy constructor?

- ▶ Dangling pointer
- ▶ Memory Leakage
- ▶ **All of the given (Page 147)**
- ▶ System crash

## Question No: 5

this pointers are not accessible for static member functions.

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► True (Page 114)

► False

## Question No: 6

A static member function cannot be declared.

► Static

► Implicit

► Explicit

► **Virtual**

## Question No: 7

\_\_\_ remain in memory even when all objects of a class have been destroyed.

► **Static variables (Page 111)**

► Instance variable

► Primitive variables

► None of given

## Question No: 8

Friend functions are \_\_\_ functions of a class.

► None of given

► object member

► **non-member (Page 136)**

► data member

## Question No: 9

\_\_\_, which means if A declares B as its friend it does NOT mean that A can access private data

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of B. It only means that B can access all data of A.

- ▶ **Friendship is one way only**
- ▶ Friendship is two way only
- ▶ NO Friendship between classes
- ▶ Any kind of friendship

## Question No: 10

The statement `objA=objB;` will cause a compiler error if the objects are of different classes.

- ▶ **True**
- ▶ False

## Question No: 11

Identify which of the following overloaded operator function's declaration is appropriate for the given call?

`Rational_number_1 + 2.325`

Where `Rational_number_1` is an object of user defined class `Rational_number`.

- ▶ `Rational_number operator+( Rational_number & obj);`
- ▶ **`Rational_number operator+(double& obj);` (Page 145)**
- ▶ `Rational_number operator+(Rational_number &obj, double& num);`
- ▶ `operator+(double& obj);`

## Question No: 12

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Which operator can not be overloaded?

- ▶ The relation operator ( >= )
- ▶ Assignment operator ( = )
- ▶ Script operator ( [] )
- ▶ **Conditional operator ( ? : ) (Page 141)**

## Question No: 13

To convert from a user-defined class to a basic type, you would most likely use

- ▶ a built-in conversion operator.
- ▶ a one-argument constructor.
- ▶ an overloaded = operator.
- ▶ **a conversion operator that's a member of the class.**

## Question No: 14

The technique in which we visualize our programming problems according to real life's problems is called

-----

- ▶ structured programming
- ▶ **object oriented Programming (Page 9)**
- ▶ procedural programming
- ▶ non of the given

## Question No: 15

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In object orientated programming, a class of objects cans \_\_\_\_ properties from another class of objects

- ▶ Utilize
- ▶ Borrow
- ▶ **Inherit**
- ▶ Adopt

## Question No: 16

A C++ class is similar to -----

- ▶ **Structure**
- ▶ Header File
- ▶ Library File
- ▶ None of the given

## Question No: 17

What problem(s) may occur when we copy objects without using deep copy constructor?

- ▶ Dangling pointer
- ▶ Memory Leakage
- ▶ **All of the given (Page 147)**
- ▶ System crash

## Question No: 18

Suppose that the Test class does not have an overloaded assignment operator. What happens when an assignment a=b; is given for two Test objects a and b?

- ▶ The automatic assignment operator is used
- ▶ **The copy constructor is used**
- ▶ Compiler error

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► Run-time error

## Question No: 19

a'A static member function can be called, even when a class is not \_\_\_\_.

- Declared
- Define
- **Instantiated** C++ How to Program, Fifth Edition(Page 871)
- Called

## Question No: 20

Identify which of the following overloaded operator function's declaration is appropriate for the given call?

Rational\_number\_1 + 2.325

Where Rational\_number\_1 is an object of user defined class Rational\_number.

- Rational\_number operator+( Rational\_number & obj);
- **Rational\_number operator+(double& obj); (Page 145)**
- Rational\_number operator+(Rational\_number &obj, double& num);
- operator+(double& obj);

## Question No:21

\_\_\_\_ provide the facility to access the data member.

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▶ **accesser function (Page 68)**

▶ private function

▶ inline function

▶ None of the given

## Question No: 22

Constant objects cannot change their state,

▶ **True (Page 105)**

▶ False

## Question No: 23

The \_\_\_\_ relationship indicates that an object contains other objects.

▶ **None of given (Page 53)**

▶ 'has-a'

▶ 'is-a'

▶ 'be-

## Question No: 24

Which one of the following features of OOP is used to derive a class from another?

▶ Encapsulation

▶ Polymorphism

▶ Data hiding

▶ **Inheritance (Page 25)**



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## Question No: 25

\_\_\_ is a relationship

► **Inheritance** (Page 25)

- Polymorphism
- abstraction
- encapsulation

## Question No: 26

\_\_\_ satisfy the condition of polymorphism

- Carbon
- Diamond
- Coal

► **all of the given** (Page 56)

## Question No: 27

A generalization-specialization relation between classes are implemented using

- data hiding
- friend classes
- encapsulation
- **inheritance** (Page 49)

## Question No: 28

The >= operator can be overloaded.

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▶ True (Page 140)

▶ False

## Question No: 29

In order to free the memory occupied by the object, we use -----

▶ Constructor

▶ Destructor (Page 92)

▶ Shallow Copy

▶ Deep Copy

## Question No: 30

Which of the following is not an example of multiple inheritances?-----

▶ Mermaid

▶ Woman (Lecture No.5)

▶ None of the given

▶ Amphibious Vehicle

## Question No: 31

Static variable can be initialized more than once.

▶ True

▶ False

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## Question No: 32

A generic class showing all the common attributes and a behavior of other classes represents a very important feature in oop called -----

▶ **Inheritance** (Page 26)

▶ Encapsulation

▶ Polymorphism

▶ Abstraction

## Question No: 33

We can get only one unique value which can be used by all the objects of that class by the use of,

▶ **static variables**

▶ dynamic variables

▶ instance variables

▶ data members

## Question No: 34

A member function having the same name as that of a class and a ~ sign with it is called,

▶ Constructor

▶ Getter

▶ Setter

▶ **Destructor** (Page 92)

## Question No: 35

Using encapsulation we can achieve

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## ► Information hiding (Page 16)

- Least interdependencies among modules
- Implementation independence
- All of given options

### Question No: 36

Inheritance is a way to

- make general classes into more specific classes. (Page 27)
- pass arguments to objects of classes.
- improve data hiding and encapsulation.
- providing class growth through natural selection.

### Question No: 37

Static variable can be initialized more than once.

- True
- False (Page 107)

### Question No: 38

For classes with common behavior, you can save effort by placing the common behavior in a \_\_\_\_.

- Derived Class
- Base class (Page 29)
- Deprived Class
- Named class

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## Question No: 39

Which of the following are an advantage of OOP?

- ▶ OOP makes it easy to re-use the code
- ▶ It provides an ability to create one user defined data type by extending the other
- ▶ It provides the facility of defining Abstract data types through which real world entities can be defined better.

▶ **All of the given options**      [Click here for detail](#)

## Question No: 40

The >= operator can be overloaded.

▶ **True (Page 140)**

▶ False

## Question No: 41

A static member function cannot be declared.

- ▶ Static
- ▶ Implicit
- ▶ Explicit
- ▶ **Virtual**

## Question No: 42

Static variables act like a global variable in the context or scope of the class.

▶ **True**

▶ False

## Question No: 43

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The compiler won't object if you overload the \* operator to perform division.

► True

► False

## Question No: 44

We can use "this" pointer in the constructor in the body and even in the initialization list of any class if we are careful,

► TRUE

► False

## Question No: 45

A C++ class is similar to -----

► Structure

► Header File

► Library File

► None of the given

## Question No: 46

Which operator can not be overloaded?

► The relation operator ( >= )

► Assignment operator ( = )

► Script operator ( [ ] )

► Conditional operator ( ? : ) (Page 141)

## Question No: 47

An overloaded operator always requires one less argument than its number of operands.

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▶ True

▶ False

## Question No: 48

A generalization-specialization relation between classes are implemented using

▶ data hiding

▶ friend classes

▶ encapsulation

▶ inheritance (Page 49)

## Question No: 49

In OOP a class is an example of \_\_\_\_

▶ Data Type

▶ Abstract Type

▶ User Defined Type (Page 66)

▶ None of the given

## Question No: 50

A class can be identified from a statement by -----

▶ Noun (Page 58)

▶ Pronoun

▶ Verb

▶ Adverb

## Question No: 51

The members of a class that can be accessed without creating the object of the class is called



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- ▶ Private member
- ▶ Data Member
- ▶ **Public Member (How to Program page 983)**
- ▶ Static

## Question No: 52

Suppose there is an object of type Person, which of the following can be considered as one of its attributes

- ▶ Name
- ▶ Age
- ▶ Work()
- ▶ **Both Name and Age (page 13)**

## Question No: 53

What a derived class can add?

- ▶ New data members
- ▶ New member functions and New friend functions
- ▶ New constructors and destructor
- ▶ **All of given**

## Question No: 54

\_\_\_\_\_ is/are used to access information hidden within an object?

- ▶ Interface
- ▶ Private data members
- ▶ **Private member functions (Page 69)**
- ▶ Both public and private members

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Question No: 55

this pointers are not accessible for static member functions.

► **True (Page 14)**

► False

Question No: 56

A static member function cannot be declared.

► Static

► Implicit

► Explicit

► **Virtual**

Question No: 57

C++ compiler does not allow to dynamically allocate memory for objects

► **False**

► True

Question No: 58

Given the following class

```
class Base{
 int Age=33;
}
```

How you can improve above class with respect to accessing the field Age?

► Define the variable Age as private

► Define the variable Age as protected

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► Define the variable Age as private and create a get method that returns it and a set method that updates it

► Define the variable Age as protected and create a set method that returns it and a get method that updates it

## Question No: 59

Friend class and friend function can be used as an alternate to each other

► True

► False

## Question No: 60

Which of the following operators always takes no argument if overloaded?

► /

► -

► +

► ++

## Question No: 61

Suppose that the Test class does not have an overloaded assignment operator. What happens when an assignment `a=b;` is given for two Test objects a and b?

► The automatic assignment operator is used

► The copy constructor is used

► Compiler error

► Run-time error

## Question No:62

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Assume a class C with objects obj1, obj2, and obj3. For the statement `obj3 = obj1 - obj2` to work correctly, if the overloaded - operator must

- ▶ take two arguments.
- ▶ **return a value**
- ▶ create a named temporary object.
- ▶ take four arguments

## Question No: 63

Which operator can not be overloaded?

- ▶ The relation operator ( `>=` )
- ▶ Assignment operator ( `=` )
- ▶ Script operator ( `[]` )
- ▶ **Conditional operator ( `? :` )** (Page 141)

## Question No: 64

We achieve independence of internal implementation from its external interface through-----

- ▶ Encapsulation
- ▶ Information Hiding
- ▶ **Abstraction**
- ▶ both encapsulation and information hiding

## Question No: 65

Which one of the following is not an object association?

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► Simple Association

► **Inheritance** (Page 49)

► Aggregation

► Composition

## **Question No: 66**

Keeping in view the principle of abstraction, which of the above information the company needs to save as employee's record?

► 2, 4

► 1, 3, 5

► **1, 2, 3** (Lecture No.3)

► 1, 2, 3, 4

## **Question No: 67**

What a derived class can add?

► New data members

► New member functions and New friend functions

► New constructors and destructor

► **All of given**

## **Question No: 68**

The concept of derived classes is involved in,

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## ► inheritance (Page 25)

- encapsulation
- structure
- array

### Question No: 69

Your chief Software designer has shown you a sketch of the new Computer parts system she is about to create. At the top of the hierarchy is a Class called Computer and under this are two child classes. One is called LinuxPC and one is called WindowsPC. The main difference between the two is that one runs the Linux operating System and the other runs the Windows System (of course another difference is that one needs constant re-booting and the other runs reliably). Under the WindowsPC are two Sub classes one called Server and one Called Workstation. How might you appraise your designers work?

- Give the goahead for further design using the current scheme
- Ask for a re-design of the hierarchy with changing the Operating System to a field rather than

### Class

- Ask for the option of WindowsPC to be removed as it will soon be obsolete
- Change the hierarchy to remove the need for the superfluous Computer Class.

### Question No: 70

Consider the code below,

```
class Fred {
public:
Fred();
```

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```
...
};

int main()
{
 Fred a[10];
 Fred* p = new Fred[10];
 ...
}
```

Select the best option,

► Fred a[10]; calls the default constructor 09 times

Fred\* p = new Fred[10]; calls the default constructor 10 times

► Produce an error

► **Fred a[10]; calls the default constructor 11 times**

**Fred\* p = new Fred[10]; calls the default constructor 11 times**

► Fred a[10]; calls the default constructor 10 times

Fred\* p = new Fred[10]; calls the default constructor 10 times

## Question No:71

Which construct is the source for the creation of an object?

► Destructor of the class

► **New operator**

► Delete operator

► Constructor of the class

## Question No: 72

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this pointers are not accessible for static member functions.

▶ **True (Page 14)**

▶ False

## Question No: 73

When a variable is define as static in a class then all object of this class,

▶ Have different copies of this variable

▶ **Have same copy of this variable**

▶ Can not access this variable

▶ None of given

## Question No: 74

\_\_\_ remain in memory even when all objects of a class have been destroyed.

▶ **Static variables (Page 111)**

▶ Instance variable

▶ Primitive variables

▶ None of given

## Question No: 75

Given the following class

```
class Base{
 int Age=33;
}
```

How you can improve above class with respect to accessing the field Age?

▶ Define the variable Age as private

▶ Define the variable Age as protected

▶ **Define the variable Age as private and create a get method that returns it and a set method that updates it**



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▶ Define the variable Age as protected and create a set method that returns it and a get method that updates it

## Question No: 76

The life of sub object is not dependant on the life of master class in \_\_\_\_.

- ▶ Composition
- ▶ **Aggregation (Page 134)**
- ▶ Separation
- ▶ None of the given

## Question No: 77

Which one is not keyword in C++?

- ▶ operator
- ▶ **B\_op**
- ▶ const
- ▶ None of given

## Question No: 78

The >= operator can be overloaded.

- ▶ **True (Page 140)**
- ▶ False

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## Question No: 79

A template provides a convenient way to make a family of

- ▶ variables and data members
- ▶ **functions and classes**
- ▶ classes and exceptions
- ▶ programs and algorithms

## Question No: 80

Which one of the following terms must relate to polymorphism?

- ▶ Static allocation
- ▶ Static typing
- ▶ **Dynamic binding (How to program , page 1103)**
- ▶ Dynamic allocation

## Question No: 81

What is true about function templates?

- ▶ The compiler generates only one copy of the function template
- ▶ **The compiler generates a copy of function respective to each type of data (Page 257)**
- ▶ The compiler can only generate copy for the int type data
- ▶ None of the given.

## Question No: 82

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Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

## ► Templates

- Overloading
- Data hiding
- Encapsulation

### Question No: 83

template <>

```
class Vector<char*> { }
```

This is an example of partial specialization.

## ► True (Page 281)

- False

### Question No: 84

Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be concrete, while classes like Sphere and Cube would normally be abstract.

- True

## ► False [Click here for Detail](#)

### Question No: 85

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A non-virtual member function is defined in a base class and overridden in a derived class; if that function is called through a base-class pointer to a derived class object, the derived-class version is used.

► True

► False

## Question No: 86

Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access

► public members of Derv.

► protected members of Derv.

► private members of Derv.

► protected members of Base.

## Question No: 87

In order to define a class template, the first line of definition must be:

► **template <typename T> (Page 281)**

► typename <template T>

► Template Class <ClassName>

► Class <Template T>

## Question No: 88

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If there is a pointer p to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a nonvirtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the \_\_\_\_ class to be executed.

- ▶ Base
- ▶ Derived
- ▶ Abstract
- ▶ **virtual (not sure)**

## Question No: 89

When the base class and the derived class have a member function with the same name, you must be more specific which function you want to call (using \_\_\_\_).

- ▶ scope resolution operator
- ▶ dot operator
- ▶ null operator
- ▶ **Operator overloading (Page 204)**

## Question No: 90

Non Template Friend functions of a class are friends of \_\_\_\_instance/s of that class.

- ▶ **All**
- ▶ One specific
- ▶ All instances of one date type
- ▶ None of the given options

## Question No: 91

The find() algorithm

- ▶ finds matching sequences of elements in two containers.
- ▶ finds a container that matches a specified container.
- ▶ **takes iterators as its first two arguments.**
- ▶ takes container elements as its first two arguments.

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## Question No: 92

If you define a vector *v* with the default constructor, and define another vector *w* with a one-argument constructor to a size of 11, and insert 3 elements into each of these vectors with `push_back()`, then the `size()` member function will return \_\_ for *v* and \_\_ for *w*.

- ▶ 11 for *v* and 3 for *w*.
- ▶ 0 for *v* and 0 for *w*.
- ▶ 0 for *v* and 3 for *w*.
- ▶ **3 for *v* and 11 for *w*.**

## Question No: 93

Which of the following may not be an integral part of an object?

- ▶ State
- ▶ Behavior
- ▶ Protected data members
- ▶ **All of given (not sure)**

## Question No: 94

Which is not the Advantage of inheritance?

- ▶ providing class growth through natural selection.
- ▶ facilitating class libraries.
- ▶ **avoiding the rewriting of code.**
- ▶ providing a useful conceptual framework

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## CS304 GRAND QUIZ

| SR NO | MCQS QUESTION                                                                                                                   | ANSWER                                                  |
|-------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 1     | In expression $c1 * c2 + c3 - c4$ which of the following will be executed in first order                                        | $C1 * c2$                                               |
| 2     | Inheritance is a way to                                                                                                         | Add features to existing classes without rewriting them |
| 3     | ----- represent "IS A" relationship                                                                                             | Inheritance                                             |
| 4     | Constructor without any parameter is called:                                                                                    | Default constructor                                     |
| 5     | Which of the following stores the address of a variable?                                                                        | Pointer                                                 |
| 6     | A Post-fix unary operator is implemented in C++ using Member function with                                                      | 1 dummy int argument                                    |
| 7     | minimum classes required in a program for implementing multiple inheritance                                                     | 3                                                       |
| 8     | Which of the following statement(s) describe constant member functions?                                                         | All of the given option                                 |
| 9     | composition is a strong relation relationship because                                                                           | Composed object cant exist independency                 |
| 10    | How many interactions are there in the following scenario Imran has the car. he drive the car and car has four wheels and doors | 3                                                       |
| 11    | if we extend our model ,and the rest of the model is not affected then it is called-----                                        | Flexibility                                             |
| 12    | choose correct declaration of overload $*=1$ operator for class complex as Member function                                      | Complex and operator $*=(const complex \& c)$           |
| 13    | Consider the following class <code>A{int a,b,c;</code>                                                                          | A,b,c                                                   |

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|    |                                                                                                                                     |                                            |
|----|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
|    | public: A():B(10),C(0)a(7) };                                                                                                       |                                            |
| 14 | Class is a blue print of                                                                                                            | Objects                                    |
| 15 | Suppose we have a class student which of the following statement is correctly declaring the pointer to the object of student class? | Student* object                            |
| 16 | In c++ composition is relationship between -----and-----objects                                                                     | Whole, part                                |
| 17 | Which of the following Depicts the proper definition of class?                                                                      | a class is description of a kind of object |
| 18 | in class attributes are represented by ---- and----- respectively                                                                   | data member, Member function               |
| 19 | which of the following represents the two Association                                                                               | Employee works for the company             |
| 20 | Entities in OO model are represented by                                                                                             | Rectangle                                  |
| 21 | Which of the following features of OOP is used to derive a class from another?                                                      | Inheritance                                |
| 22 | In C ++ which of the following is defined as a stream insertion operator?                                                           | <<                                         |
| 23 | choose the correct declaration of overloaded equality (==)operator for class string as a Member function                            | Bool operator==(const string and s)        |
| 24 | ---- is the relationship between the part object and the whole object                                                               | Composition                                |
| 25 | object hides its information from others this concept is known as                                                                   | information hiding                         |
| 26 | the concept of derived classes is involved in                                                                                       | Inheritance                                |
| 27 | abstraction provides information according to                                                                                       | user perspective                           |
| 28 | which of the following is not an object Association                                                                                 | Inheritance                                |
| 29 | which of the following classes are used by amphibious vehicle to inherit                                                            | both land and water vehicle                |



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|    |                                                                                                             |                                                               |
|----|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
|    | characteristics                                                                                             |                                                               |
| 29 | object data includes                                                                                        | both states and attributes of an object                       |
| 30 | choose correct declaration of overloaded inequality operator for class string as non member friend function | friend pool operator !<br>=(const string &s1,const string &2) |
| 31 | How many objects of a given class may be constructed in an application?                                     | as many as the application ask for                            |
| 32 | how many objects of a given class may be constructed in an application                                      | as many as the application ask for                            |
| 33 | Which of the following statements describe constant member functions?                                       | all of the given option                                       |
| 34 | student *ptr=new student(" Ali"); is an example of                                                          | static memory allocation                                      |
| 35 | in----- base class can be replaced by the derived class                                                     | extension                                                     |
| 36 | Which of the following is a correct way of declaring constant variable ROLLNO?                              | cosnst int Roll no;                                           |
| 37 | which of the following operator doesn't take argument hit overloaded                                        | none of the given choices                                     |
| 38 | which of the following is part of the Class diagram                                                         | all of the given option                                       |
| 39 | a child inherits characteristic from its                                                                    | parent                                                        |
| 40 | which of the following issues can be caused by the use of friend function                                   | programming Bugs                                              |
| 41 | which of the following is not true about constant number function                                           | all of the given option                                       |
| 42 | Which one is not the main features of object oriented programming?                                          | exception handling                                            |
| 43 |                                                                                                             |                                                               |
| 44 | in object oriented programming, objects communicate with each other through                                 | messages                                                      |
| 45 | static data member is declared                                                                              | inside the class                                              |
| 46 | the concept of derived classes is involved                                                                  | inheritance                                                   |

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|    |                                                                                                                                               |                                             |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
|    | in                                                                                                                                            |                                             |
| 47 | Which of the following can we attribute of an object "Ali"?                                                                                   | address                                     |
| 48 | Which of the following keyword tell the compiler to substitute the code within the function definition for every instance of a function call? | inline                                      |
| 49 | inheritance is the relationship between                                                                                                       | classes                                     |
| 50 | how many objects are involved in the binary Association                                                                                       | 2                                           |
| 51 | all objects of same class has ----- attributes                                                                                                | same                                        |
| 52 | which of the following concept is more close to encapsulation                                                                                 | information hiding                          |
| 53 | member initialization list is used to                                                                                                         | initialize data member                      |
| 54 | aggregation between objects show that                                                                                                         | objects have weak relationship              |
| 55 | there is /are ways to create user defined types for objects in C ++                                                                           | two                                         |
| 56 | composition is a strong relationship because:                                                                                                 | composed object can not exist independently |
| 57 | in polymorphism messenger can be interpreted in different ways depending upon the----- class                                                  | receiver                                    |
| 58 | which of the following is most suitable advantage of information hiding                                                                       | simplifies the model                        |
| 59 | the process of extracting common characteristics from two or More classes and combining them into a generalized superclass is called          | Generalization                              |
| 60 | ----- operator is used to create objects at run time                                                                                          | new                                         |
| 61 | choose the correct option to declare an overloaded Stream insertion operator for                                                              | friend ostream and operator<<(ostream       |

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|    |                                                                                                                    |                               |
|----|--------------------------------------------------------------------------------------------------------------------|-------------------------------|
|    | userdefined class "string" as a non-member friend function                                                         | and os,const string & s)      |
| 62 | static data member is declared                                                                                     | inside the class              |
| 63 | an instance of user defined type is called                                                                         | object                        |
| 64 | which of the following is true about constructor                                                                   | all of the given option       |
| 65 | Which of the following correct way of declaring constant variable ROLLNO?                                          | const int ROLLNO;             |
| 66 | inheritance is shown by                                                                                            | filled arrow                  |
| 67 | which of the following operators operate on one operand                                                            | unary operator                |
| 68 | which of the following will happen when a constant function tried to change the value of data members of the class | compile time error will occur |
| 69 | in object oriented programming, objects communicate with each other through---                                     | messages                      |
| 70 | How the information hidden within an object can be accessed?                                                       | through its interface         |
| 71 | object orientation approach can be achieved using object and their-----                                            | interactions                  |
| 72 | constructors have ----- return type                                                                                | no                            |
| 73 | which of the following is an example of unary operator                                                             | increment operator(++)        |
| 74 | memory is allocated to non-static members only when                                                                | object is created             |
| 75 | in class B inherit from class A which of the following is child class                                              | class B                       |
| 76 | the----- tells the compiler what task the function will be performing                                              | function definition           |
| 77 | Consider the statement" room has chair" which of the following type of Association exists between whom and chair?  | aggregation                   |
| 78 | how many objects are involved in the N-                                                                            | more than 3                   |

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|    |                                                                                                           |                                                      |
|----|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------|
|    | ary Association                                                                                           |                                                      |
| 79 | deconstruct is a function which has the same name as that of class but starts with a----- sign            | ~                                                    |
| 80 | which of the following is a tangible entity                                                               | car                                                  |
| 81 | member initializer list is used to                                                                        | initialize data members                              |
| 82 | we can get the address of variable stored in pointer using-----                                           | & symbol                                             |
| 83 | two sentence object oriented programming book in bookshelf is an example of                               | aggregation                                          |
| 84 | which of the following best defines the obstruction                                                       | hide the implementation and show current perspective |
| 85 | which of the following define default constructor 5 class student                                         | student(){//..}                                      |
| 86 | object data includes                                                                                      | both state and attributes of an object               |
| 87 | minimum classes required in a program for implementing multiple inheritance                               | 3                                                    |
| 88 | information hiding can be achieved through-----                                                           | encapsulation, abstraction                           |
| 89 | compiler generated defaulter structure is called----- hand user written default constructor is called---- | implicit, explicit                                   |
| 90 | which of the following is true about constructor                                                          | all of the given option                              |
| 91 | In which of the following different forms office single entity exist?                                     | polymorphism                                         |
| 92 | data members of a class student can be:                                                                   | all of the given option                              |
| 93 | the----- turn off the compiler what does the function will be performing                                  | function definition                                  |
| 94 | Suppose a class does not have any                                                                         | compiler will call                                   |

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|     |                                                                                                                                              |                                                                                                  |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
|     | constructor what will happen when an object of this class is created?                                                                        | implicit default constructor                                                                     |
| 95  | how can we differentiate between constructor and destructor                                                                                  | destructor are preceded with a Tilda(~) symbol, and constructor are not preceded with any symbol |
| 96  | ---- is used for hiding the details of an object from the other parts of a program                                                           | encapsulation                                                                                    |
| 97  | Class diagram include                                                                                                                        | class name, attributes, operation                                                                |
| 98  | through interface was access objects-----                                                                                                    | behavior                                                                                         |
| 99  | composition is a strong relationship because                                                                                                 | compose object cannot exist independently                                                        |
| 100 | copy constructor is called when:                                                                                                             | an object is created in term of Preexisting object                                               |
| 101 | choose the correct option to declare an overloaded stream insertion operator for a user defined class string as a non-Member friend function | friend ostream & operator(ostream &os.const string &s)                                           |
| 102 | a post-fix unary operator implemented in C plus plus using non Member function with---- arguments                                            | two                                                                                              |
| 103 | in class attributes and behaviors are represented by---- and--- respectively                                                                 | data members, member functions                                                                   |
| 104 | which of the following best describes the relationship between book and book chapter classes                                                 | composition                                                                                      |
| 105 | which of the following is true about destructor                                                                                              | all of the given option                                                                          |
| 106 | in composition----- are called from composing objects to composed subjects                                                                   | destructors                                                                                      |
| 107 | which of the following can be used an                                                                                                        | Global variables                                                                                 |

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|     |                                                                                                                  |                                            |
|-----|------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
|     | alternative to Static members                                                                                    |                                            |
| 108 | which of the following statement is true about static function of a class                                        | it is used to access static data members   |
| 109 | which of the following operator does not take any arguments if overloaded                                        | none of the given choices                  |
| 110 | what is the output of the following code<br><code>int main(){int cost x=10;cout&lt;&lt;+<br/>+x;return0;}</code> | error                                      |
| 111 | abstraction includes-----                                                                                        | relevant information                       |
| 112 | Which of the following can be the behavior of an object "Usman"?                                                 | eat                                        |
| 113 | a class defined within another class is                                                                          | nested class                               |
| 114 | when an object is initialized, is automatically called by the compiler                                           | constructor                                |
| 115 | the compiler treats the Member function of a class as                                                            | inline function                            |
| 116 | an object has attributes operation ,and-----                                                                     | unique identity                            |
| 117 | Which of the following is not an access specifier in C ++?                                                       | hidden                                     |
| 118 | which of the following parts of an object exhibits its state                                                     | data                                       |
| 119 | ---- is the relationship between the part object and the whole object                                            | composition                                |
| 120 | how can we identify classes from a given problem statement                                                       | by extracting nouns from problem statement |
| 121 | what "friend" should be placed in                                                                                | function declaration                       |
| 122 | generalization is----- approach                                                                                  | bottom-up                                  |
| 123 | static data members can be accessed through-----                                                                 | dot operator and scope resolution operator |
| 124 | composition age----- and aggregation is----- relationship                                                        | strong, weak                               |
| 125 | suppose there is an object of type                                                                               | name                                       |

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|     |                                                                                                                                                                |                                                               |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
|     | person, which of the following can be considered as one of its attributes                                                                                      |                                                               |
| 126 | Which of the following depicts the proper definition of class?                                                                                                 | a class is a description of a kind of object                  |
| 127 | a child inherits characteristics from its-----                                                                                                                 | parent                                                        |
| 128 | the car is composed of wheels what is the relationship between car and wheels?                                                                                 | composition                                                   |
| 129 | ----- is creating objects of one class inside another class                                                                                                    | composition                                                   |
| 130 | the other name of specialization is-----                                                                                                                       | restriction                                                   |
| 131 | suppose a is an int type static data member class test then which of the following is correct way of initialization the static variable outside the class test | int test::a=0                                                 |
| 132 | Which of the following stores the address of variable?                                                                                                         | Pointer                                                       |
| 133 |                                                                                                                                                                |                                                               |
| 134 |                                                                                                                                                                |                                                               |
| 135 | static data member is declared                                                                                                                                 | inside the class                                              |
| 136 | polymorphism makes the system                                                                                                                                  | all of the given option                                       |
| 137 | which of the following statement best describes the constructor                                                                                                | constructor is used to initialize the data members of a class |
| 138 | data member(s) of a class student can be                                                                                                                       | all of the given option                                       |
| 139 | which of the following is a strong relationship                                                                                                                | composition                                                   |
| 140 | all objects of same class has----- attributes                                                                                                                  | same                                                          |
| 141 | in C ++ composition is a relationship between----and---- objects                                                                                               | whole, part                                                   |
| 142 | suppose 12 class string assignment operator is overloaded with following                                                                                       | compiler will generate compile time error                     |

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|     |                                                                                                                                                                                |                                                           |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
|     | declaration void operator=(const string &); what will happen when we will write following statement in main()<br>str=str2=str3;where,str1.str2,str3 are object of class string |                                                           |
| 143 | in class B inherits from class A then it contains all characteristics of----                                                                                                   | class A                                                   |
| 144 | which of the following is not an example of multiple inheritance                                                                                                               | car                                                       |
| 145 | which of the following can be created without any object of a class?                                                                                                           | static data members                                       |
| 146 | consider the following code segment<br>class test{int a;int b;int c;}                                                                                                          | junk value,5, junk value                                  |
| 147 | object orientation approach can be achieved using objects and their-----                                                                                                       | interactions                                              |
| 148 | member functions defined inside a class declaration are----- by default                                                                                                        | private                                                   |
| 149 | how can we identify classes from a given problem statement                                                                                                                     | by extracting nouns from problem statement                |
| 150 | which of the following statement is not true about static variable of a class?                                                                                                 | static variable belongs to particular instance of a class |
| 151 | there are----- types of constructor in C++                                                                                                                                     | 3                                                         |
| 152 | statement yasir is a friend of Ali and Ali is a friend of yasir is an example of                                                                                               | two way Association                                       |
| 153 | this pointer cannot be passed implicitly to----- functions                                                                                                                     | static member                                             |
| 154 | which of the following issues can be caused by the use of friend function                                                                                                      | all of the given                                          |
| 155 | in----- send a message does not need to know exact class of the receiver                                                                                                       | none of the given option                                  |



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|     |                                                                                                                         |                                           |
|-----|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| 156 | minimum classes required in a program for implementing multiple inheritance                                             | 3                                         |
| 157 | -----Associates the object of the exact two classes                                                                     | binary Association                        |
| 158 | main characteristics of information hiding is                                                                           | keep the model simple                     |
| 159 | static data member is declared                                                                                          | inside the class                          |
| 160 | in expression $b=c/d$ , the arity of division(/) operator is                                                            | 2                                         |
| 161 | we can allocate dynamic memory in C ++ through operator                                                                 | new                                       |
| 162 | Which of the following is true about accessor functions?                                                                | all of the given option                   |
| 163 | in---- base class cannot always be replaced by the derived class                                                        | specialization                            |
| 164 | abstraction help to                                                                                                     | reduce the complexity                     |
| 165 | relationship in which child object gets destroyed if parent object is destroyed                                         | composition                               |
| 166 | object oriented programming is better than procedural programming because:                                              | it is more close to reality               |
| 167 | The derived class inherits behavior from the base class and also exhibits its own behavior. this concept is known as--- | Overriding                                |
| 168 | the other name of subtyping is                                                                                          | Extension                                 |
| 169 | A good model is----- related to life problem                                                                            | closely                                   |
| 170 | in generalization general class represents the ----- behavior of the class                                              | common                                    |
| 171 | how can we identify from a given problem statement?                                                                     | by extracting noun from problem statement |
| 172 | which of the following is not a access specifier in C ++                                                                | hidden                                    |
| 173 | in composition----- are called from composing objects to composed objects                                               | destructors                               |

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|     |                                                                                                                                                                                |                                                           |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| 174 | a class have only---- destructor                                                                                                                                               | one                                                       |
| 175 | suppose you have following C + + statement<br>int old value=10; int new value=++old value; what will be the value of old value and newValue after executing above statements ? | oldVlue=11,newVlue=11                                     |
| 176 | in constant Member function the type of this pointer is                                                                                                                        | constant pointer to constant data                         |
| 177 | ----- is creating objects of one class inside another class                                                                                                                    | composition                                               |
| 178 | ----- Associates the objects of the exact three classes                                                                                                                        | ternary Association                                       |
| 179 | generalization is implemented through                                                                                                                                          | inheritance                                               |
| 180 | which of the following is strong relationship                                                                                                                                  | composition                                               |
| 181 | composition is a strong relationship because                                                                                                                                   | composed object cannot exist independently                |
| 182 | ----- is set of function of an object exposed of other objects                                                                                                                 | interface                                                 |
| 183 | suppose student is a class, which of the following defines constructor with one parameter for class student                                                                    | student(char* aName);                                     |
| 184 | which of the following is not true about constant Member function                                                                                                              | all of the given option                                   |
| 185 | ----- define the order of evaluation of an operator is an expression                                                                                                           | operator precedence                                       |
| 186 | choose correct declaration of overloaded inequality(!=) operator for class string as non member friend function                                                                | friend bool operator!=(const string &s1,const string &s2) |
| 187 | which of the following is most suitable advantage of information hiding                                                                                                        | simplifies the model                                      |
| 188 | a fan has wings, which type of relation exists between fan and Wings in this                                                                                                   | composition                                               |

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|     |                                                                                                         |                                                                                         |
|-----|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
|     | sentence                                                                                                |                                                                                         |
| 189 | multiple inheritance is implemented between class through----- symbol                                   | comma(,)                                                                                |
| 190 | postfix unary operator is implemented in C ++ using non Member function with arguments                  | two                                                                                     |
| 191 | Supposed derived class is inherited from base class what happens when a derived class object is created | the constructor of base class exact is executed before the constructor of derived class |
| 192 | assignment operator are                                                                                 | right associative                                                                       |
| 193 | a postfix unary operator is implemented in C ++ using number function with                              | 1 dummy int argument                                                                    |
| 194 | which of the following is true about exception functions                                                | all of the given option                                                                 |